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Master's Programme in People Management and Organizational Development

Development of safety culture in safety-critical business networks

Experiences and key practices from Finnish energy and industrial sectors

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Abstract

This thesis aims to explore how safety culture can be developed within safety-critical business networks, where inadequate management of risks could harm people, environment, or property. The emphasis on ensuring safety varies by industry, but studies on major industrial projects have indicated that schedule, quality, and financial challenges often arise from an unsatisfactory safety culture. Additionally, the evolution of management models and increased outsourcing have made coordination between organizations with varying cultures, goals, and processes more challenging.

The theoretical framework encompasses organization culture, safety culture, and organizational development intervention theories commonly applied in the industry. The focus is given on appearance and development of safety culture in the organizational and business network perspectives. Based on these key theories and other industry literature, a qualitative research was conducted to explore the practices and factors influencing safety culture development in organizational collaboration. The empirical data was collected through interviews with managers and leaders in organizations within safety-critical industries.

The findings of this research align with the reviewed literature, conforming that strong safety culture forms a solid foundation to support achieving strategic goals. A wide range of practices for influencing elements of safety culture were identified, with an emphasis on forming a unified understanding of shared goals and creating common processes for collaboration and communication. Top management commitment was identified as the most critical factor in the development of safety culture, often requiring strong financial incentives. These incentives can be implemented in internal reward systems or externally through procurement activities or contractual criteria.

Based on the theoretical framework and research findings, implementation actions for safety culture development were developed for both individual organizations and business networks. Additionally, for future considerations, safety culture related aspects could be included in corporate sustainability reporting or added as criteria for investment financing, providing strong external incentives for organizations to consider safety aspects of their operations and to develop their safety culture in a more sustainable manner.

Keywords safety culture, organization culture, safety-critical industry, business network, leadership and management for safety, organization development, change management

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Tiivistelmä

Tämän pro gradu -tutkielman tavoitteena on tutkia, miten turvallisuuskulttuuria voidaan kehittää turvallisuuskriittisissä liiketoimintaverkostoissa, joissa riskien riittämätön hallinta voi vahingoittaa ihmisiä, ympäristöä tai omaisuutta. Turvallisuuden varmistamisen painotus vaihtelee toimialoittain, mutta tutkimukset suurista teollisuus-projekteista ovat osoittaneet, että ilmenevät aikataulu-, laatu- ja taloushaasteet johtuvat usein puutteellisesta turvallisuuskulttuurista. Lisäksi johtamismallien kehitys ja lisääntynyt ulkoistaminen ovat tehneet erilaisten kulttuurien, tavoitteiden ja prosessien omaavien organisaatioiden välisen toiminnan koordinoinnista haastavampaa.

Tutkimuksen teoreettinen viitekehys kattaa toimialoilla yleisesti sovelletut organisaatio- ja turvallisuuskulttuurin sekä organisaation kehittämistoimenpiteiden teoriat, keskittyen turvallisuuskulttuurin ilmenemiseen ja kehittymiseen organisaatioissa sekä liiketoimintaverkostoissa. Näihin teorioihin ja muuhun toimialakirjallisuuteen perustuen toteutettiin laadullinen tutkimus, jossa tutkittiin turvallisuuskulttuurin kehittämiseen vaikuttavia käytäntöjä ja tekijöitä organisaatioiden välisessä yhteistyössä. Empiirinen aineisto kerättiin haastattelemalla turvallisuuskriittisten toimialojen organisaatioiden päälliköitä ja johtajia.

Tutkimuksen tulokset vastaavat tarkastellun kirjallisuuden näkemystä siitä, että vahva turvallisuuskulttuuri muodostaa hyvän perustan strategisten tavoitteiden saavuttamiselle. Tutkimuksessa tunnistettiin laaja joukko käytäntöjä, joilla turvallisuuskulttuuriin voidaan vaikuttaa, joista yhtenäisen tavoitekäsityksen muodostaminen sekä yhteisten prosessien luominen yhteistyöhön ja viestintään korostuivat eniten. Ylimmän johdon sitouttaminen korostui tärkeimpänä tekijänä turvallisuuskulttuurin kehittämisessä, mikä usein vaatii vahvoja taloudellisia kannustimia. Näitä voidaan luoda sisäisesti organisaation palkitsemisjärjestelmissä, tai ulkoisesti osana hankintatoimia tai yhteistyösopimuksia.

Teoreettisen viitekehyksen ja tutkimustulosten pohjalta luotiin toteutusehdotuksia turvallisuuskulttuurin kehittämiseksi sekä yksittäisen organisaation että liiketoiminta-verkostojen näkökulmasta. Lisäksi huomioitiin, että turvallisuuskulttuuriin liittyviä näkökohtia voidaan tulevaisuudessa sisällyttää osaksi yritysten kestävyysraportointia tai lisätä investointirahoituksen kriteereiksi, mikä loisi vahvoja ulkoisia kannustimia organisaatioille huomioida liiketoimintansa turvallisuusnäkökohtia ja kehittää turvallisuuskulttuuria kestävämmällä tavalla.

Avainsanat turvallisuuskulttuuri, organisaatiokulttuuri, turvallisuuskriittinen toimiala, liiketoimintaverkosto, turvallisuusjohtaminen, organisaatiokehitys, muutosjohtaminen

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1 Introduction

1.1 Background and research problem

Safety-critical organizations include all those whose operations involve hazards that could cause harm to people, environment, or property if inadequately managed. These organizations exist across various industries, where different tasks and operational activities could cause significant risks. For example, aviation, nuclear energy, chemical industry, and rail transport are traditionally considered as safety-critical industry sectors. The emphasis on ensuring safety varies depending on the industry as they face varying risks and have different traditions, leading to different safety priorities. (Reiman & Oedewald, 2008, p. 4).

To ensure public safety, safety-critical industries are subject to regulation by governmental agencies, making compliance with these external regulations essential for organizations in safety-critical environments. Regulatory oversight covers all aspects from operating permits to detailed operational procedures, aiming to safeguard employees and society. (Reiman & Oedewald, 2008, pp. 28-30). However, in recent years, there has been a shift away from prescriptive regulation toward goal-oriented legislation which encourages self-regulation, granting safety-critical organizations freedom and responsibility to manage their operations within broadly defined safety goals (Grote, 2018, p. 637). For example, in Finland there is an ongoing regulatory renewal in the nuclear energy sector with objectives to make regulation and guidelines more goal-oriented and risk-conscious (STUK, n.d.), and to emphasize the responsibilities of business operators' (FinNuclear, 2024). Furthermore, standardization of hydrogen production is progressing towards developing a common European framework to support the expanding hydrogen industry (SFS, n.d.; Tukes, 2024). Such legislative developments allow organizations to proactively participate in and influence safety management aspects, rather than just reacting to external changes. This approach makes a significant shift for safety-critical organizations, as compliance will be determined not only by detailed instructions on operations, but by having acceptable plans for achieving sustainable safety performance and capabilities to execute them (Grote, 2018, pp. 637-638).

Management and organizational models are also continuously evolving where network-based operations and flat organization models emphasizing innovation and learning have been preferred in recent times. Simultaneously, the use of outsourcing and temporary labor has increased. (Cummings et al., 2019, pp. 585-589). These have raised concerns whether conventional business management theories are applicable to safety-critical

organizations as they are based on fundamentally different principles than what safety research and development have traditionally considered (Reiman et al., 2015; Gotcheva et al., 2017). Classical safety thinking tends to be conservative and mainly suitable for stable conditions, whereas innovative thinking prefers taking risks and may see safety aspects as a hindrance to agile progress or even contradicting value to profitability. Furthermore, the increased outsourcing and complexity in technology and organization structures provide additional challenges for safety-critical industries (Milch & Laumann, 2016) as industrial projects are often carried out by networks of organizations, where some participants may not have extensive experience from that specific industry and knowledge of their practices (Gotcheva, 2016). These organizations may also differ by their cultures, goals and operational processes which makes coordinating activities between stakeholders more challenging (Oedewald & Gotcheva, 2015a). For instance, safety might not always be the first and most important guiding value (Macchi et al., 2013).

Evidence from many major industrial projects indicates that they have experienced schedule, quality and financial challenges in which unsatisfactory safety culture has been recognized as one contributing factor to them (Ahola et al., 2014; Reiman & Viitanen, 2018; Ruuska et al., 2011). As these business-related challenges often cause only financial problems, they may have been often neglected as '*unfortunate characteristics of project business*', even though they could reflect deeper potential organizational issues which could influence their safety performance in the future (Gotcheva, 2016). Therefore, authorities in the nuclear industry, and gradually in other safety-critical industries as well, have begun to demand good safety culture and its demonstration from industrial organizations. The recognition and creation of safety culture concept aimed to highlight that factors related to leadership and other human aspects of organization can influence the occurrence of accidents and other events. Previously, these events might have been classified as technical failures, human errors, or individual negligence. (Reiman & Oedewald, 2008, pp. 61-62).

However, many safety culture development efforts are typically targeted on working practices or changing the mindset of employees who do the practical work instead of at management of the organization (Viitanen, 2015, p. 40). This can be problematic because culture is not limited to the lower-level employees but rather it is ingrained into all levels of the organization and its operations (Williams, 2017, pp. 59-60). Additionally, the focus is often given on occupational aspects of safety rather than overall safety impacts (Oedewald & Gotcheva, 2015b, p. 108). While there may be some overlap between the factors influencing organizational performance in these different safety aspects, the risks are

fundamentally caused by different mechanisms. For example, occupational safety risks, such as lack of protective gear or dangerous working conditions, directly impact individual employees and can be relatively easy to observe and avoid with common sense. In contrast, safety risks of industrial processes are more often caused by systemic failures, making them more difficult to observe and mitigate due to their sociotechnical complexity. (Viitanen, 2015, pp. 45-46). Therefore, management for safety should emphasize holistic leadership within organizations, where organizational excellence, employee well-being, and overall safety are equally valued objectives (Reiman & Oedewald, 2008, pp. 207-208).

1.2 Research objective and questions

As new organizations expand into safety-critical industries, developing a sustainable safety culture is essential to mitigate risks and to ensure safety of their operations (Oedewald & Gotcheva, 2015a). This includes also all vendors, service providers, and other partner organizations in their business networks, as their safety performance impacts the entire industry throughout the lifecycle of their operations (Oedewald & Gotcheva, 2015b, pp. 116-117). Therefore, the primary objective of this study is to explore ***how safety culture can be developed within safety-critical business networks***. This broad research theme can be divided into following research questions:

- How does safety culture appear in business networks, and how is it assessed?
- How elements of safety culture can be influenced in practice?
- Which factors contribute to the development of safety culture?

The research questions are primarily descriptive in nature to provide a comprehensive understanding of the experiences in the development of safety culture within safety-critical industries. Descriptive questions help map out the phenomena, identify effective influencing practices, and determine factors that could explain these challenges. This approach systematically analyzes existing conditions and methods to generate insights that are both specific and actionable.

The outcomes of this study are intended to be considered by the management of individual organizations that are already operating in or considering expanding into safety-critical industries. By examining academic and industrial literature together with insights from industry representatives, this study aims to develop actionable suggestions on how to promote a sustainable safety culture that can enhance inter-organizational collaboration and contribute to the overall safety development of networked organizations.

2 Theoretical framework

This study focuses on the appearance and development of safety culture at the organizational and business network levels, especially in collaboration across organizational boundaries. Since both levels consist of groups of individuals and some phenomena may be similar between intra-organization and inter-organization contexts, their perspectives and roles are also briefly discussed. Therefore, the main theoretical framework of this study considers the following aspects:

- Formation and changing of organization culture
- Safety culture and its development
- Development interventions from single and multi-organization perspectives

2.1 Organization culture

The concept of organizational culture has been a subject of considerable academic debate for multiple decades with various approaches to study and define including Schein (1985; 2017), Hofstede (1991; 2010) and Weick (1987; 1998). In this context, culture can be divided into multiple categories or layers including national, occupational, and organizational layers (Hofstede et al., 2010, pp. 17-18). Cultural influence on people is often referred to as the learned conscious and unconscious ways in which they think, feel, and behave as members of certain socialized entities (Williams, 2017, pp. 59-60). While differences between cultures are easily recognized at the national level, they are often more challenging to identify at the occupational or organizational levels (Schein, 2010, pp. 7-8; 2017, p. 81). Studies on cultural differences have found that they tend to make the management of organizational activities more complex and challenging (Williams, 2017, p. 171).

2.1.1 Definition of organization culture

One of the most well-known models of organization culture is Edgar Schein's (1985; 2017) depth model of culture which has been widely utilized in organizational development interventions (Cummings et al., 2019, pp. 552-558; Williams, 2017, p.63), and it has also served as an underlying view for many safety culture models and assessment methods (Reiman & Oedewald, 2008, p. 62; IAEA, 2002b, pp. 3-10; IAEA, 2016c, p. 4; Reiman & Rollenhagen, 2018, pp. 659-660). Schein (2017, p. 6) himself has defined organization culture as:

"The culture of a group can be defined as the accumulated shared learning of that groups as it solves its problems of external adaption and internal integration; which has worked well enough to be considered valid and, therefore, to be taught to new members as the correct way to perceive, think, feel, and behave in relation to those problems. This accumulated learning is a pattern or system of beliefs, values, and behavioral norms that come to be taken for granted as basic assumptions and eventually drop out of awareness."

When trying to examine a culture of an organization, three different levels of cultural elements can be recognized based on how visible they are to the observer. These levels, as shown in Figure 1, range from very clearly sensible artifacts, to espoused beliefs and values, and into something lying deep in the organizations' cultural essence which are described as underlying assumptions (Schein, 2017, pp. 17-18). The circular pattern of arrows indicates that artifacts are shaped by the lower levels, but it can also work in other way around, where artifacts influence values and eventually underlying assumptions.

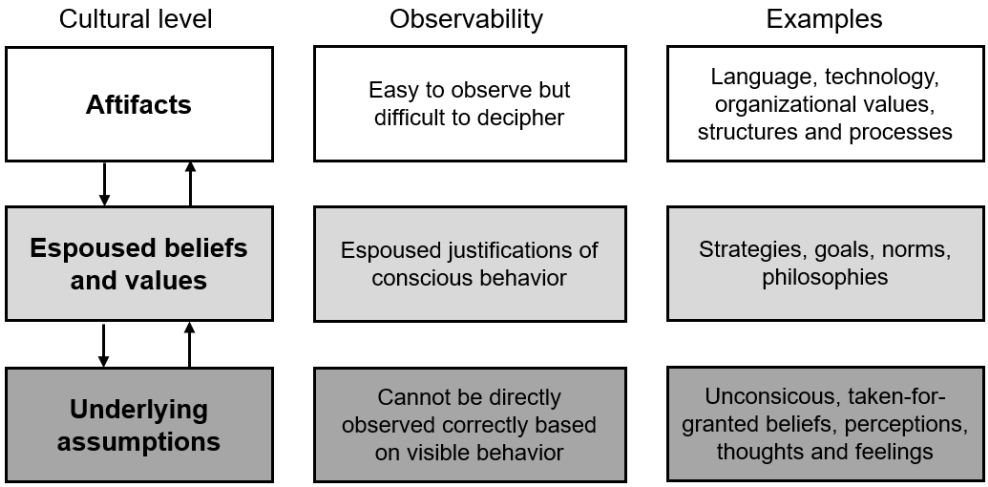


Figure 1. Three levels of organizational culture (Schein, 2017, p. 18; Reiman et al., 2008).

On the first surface level are all the cultural elements that one can easily see, hear and feel when interacting with an unfamiliar culture. These include for example logos, published list of values, physical environment such as office, dress codes, and behavior of employees of the organization. All organizational structures, charts, and workings processes that create such behavior are also included in these artifacts. Even though these artifacts can be easily described by an external observer, it is difficult to decipher what they mean for its members, or whether they even reflect important underlying assumptions of a given group. (Schein, 2017, pp. 17-18; Reiman et al., 2008, pp. 10-11).

On the second level are the espoused beliefs and values which are consciously expressed by the members of the group. They cannot be seen but are recognized when observing what people say and how decisions are made or explained. These are the ways to think, act and behave that the group has validated to be their own operational rules of behavior. Espoused beliefs and values can be accurately articulated and trained to new members of the group as they serve as normative guides on how to behave or act in certain situations. These organizational norms can predict the observable behavior at the artifacts level but if they are not based on prior learning, they may also only predict what people will say in certain situations which is different how they will act in the end. (Schein, 2017, pp. 19-21; Reiman et al., 2008, pp. 10-11).

On the third, and the deepest level, are the unconsciously held underlying assumptions and beliefs, which are the unwritten rules of the organization that are taken for granted. They are so widely spread across the organization and have such a strong hold that they are rarely discussed or thought about unless someone tries to change or violate them. This makes underlying assumptions and beliefs difficult to change as it would destabilize the group's cultural core and as humans naturally seek stability and comfort. Based on espoused beliefs and values, as well as artifacts, it is difficult for an outsider to interpret the underlying assumptions. Espoused beliefs and values are not always aligned with the underlying assumptions as they may represent more of a desired state or ideal. (Schein, 2017, pp. 21-25; Reiman et al., 2008, pp. 10-11).

The main point of view that Schein (2017, pp. 10-11) adds on top of many other concepts of organization culture is that in addition to shared norms, values, behavior patterns, rituals and traditions, the culture adds also elements of structural stability, depth, breadth and integration. In here, the structural stability implies that culture is stable and hard to change because it provides identity, meaning and predictability which defines the group. Depth and breadth imply that culture is usually the deepest part of a group which makes it less visible than other parts, and when the culture has developed it covers all of group's operations. These elements are integrated together to form a culture of a group with certain values, behaviors, and climate to help them deal with its primary tasks and internal operations in various environments. (Reiman & Rollenhagen, 2018, p. 660). Although Schein's (2017) model emphasizes underlying assumptions as the core of culture and the source of behavior, it is essential to understand that behavior is influenced by multiple complex factors. These include situational elements, such as time pressure, and individual factors, such as personality, emotions, fatigue, and abilities (Reiman et al., 2008, p. 11).

2.1.2 Subcultures within a culture

While it might be tempting to assume that organizations develop only one shared and consistent way of thinking and behaving, the reality is more complex. Schein (2010, pp. 55-57; 2017, p. 221) notes that there could grow powerful subcultures which reflect the overall culture of an organization but also differ in important ways. This formation of subcultures in smaller groups of an organization is essential to be understood as their interaction will have an impact on how the overall culture of the organization may evolve over time. Subcultures can often form in organizations around occupational and functional groups, and according to hierarchical levels of an organization, where group members often share similar educational backgrounds or organizational experiences. Groups develop their own basic assumptions, language, and ways of thinking and behaving, which will eventually lead into a situation often referred to as '*silos*'. (Schein, 1996, p. 12; 2010, p. 55). Additionally, they likely have different perspectives on the organization's core mission, execution methods, and their own roles in it (Reiman, 2009, pp. 15-17). Schein (1996, pp. 13-16; 2017, p. 221) divides the culture formed in the organization into three main subculture categories:

- Operator culture (organization based)
- Engineering culture (global occupational community)
- Executive culture (global occupational community)

In here, the operator culture covers all groups and people who get the practical work done within the organization. Their shared assumptions arise from the line units of the organization as it attempts to operate efficiently and safely. Therefore, the operator culture is primarily shaped by shared organizational experiences and is typically the culture referred to when discussing an organization's overall culture. (Schein, 1996, p. 13; 2017, pp. 221-223; Reiman, 2009, p. 17).

The engineering culture covers all groups and people responsible for designing basic technology, processes, and implementation strategies as they are more concerned about innovation, improvement, and redesign. This culture is most visible in traditional engineering functions, but it includes also other designing and implementing occupational communities whose shared assumptions are based on common education, work experience, job requirements, and external occupational reference group. (Schein, 1996, p. 14; 2017, pp. 224-225; Reiman, 2009, pp. 15-17).

The executive culture covers the top leaders and their closest subordinates whose worldview and basic assumptions are built around financial survival and growth of the organization with aim to minimize costs and maximize profits. As individuals rise in the hierarchy, they become more distant to the operations of the organization as the units they manage grow larger and they cannot manage all the people directly. This makes it more likely that their way of thinking and worldview gradually diverges from the operators, and therefore they need to develop routines and rules to manage the organization. (Schein, 1996, p. 15; 2017, pp. 226-228; Reiman, 2009, p. 15).

In large organizations the size of variations among subcultures can be remarkable which can be often found out as communication challenges between separate departments of the organization, or even lead into tensions and conflicts between them. Thus, for example making a cross-functional project team work effectively together is challenging as its members bring their own cultural background into the project. This can complicate information exchange and achieving mutual understanding as communication needs to be put into the appropriate language or format for other cultures to understand it, often resulting in translations that may distort the initial information and intention. (Schein, 1996, p. 12; 2017, pp. 119-120; Reiman, 2009, pp. 15-17).

2.1.3 Formation of organization culture

Organization culture can emerge in two main processes depending on the formality and structure of the group. In unformal and unstructured groups spontaneous interactions create behavioral norms after a certain time as its members start to have common history of shared learning. However, not every collection of people will develop a culture as it will require a stable membership and some group bonding events that will make them a group with a history of shared experiences. (Schein, 2017, pp. 127-130).

In more formal and structured groups an individual creates the group or becomes its leader. This could happen for example when an entrepreneur starts a new company, or someone is selected to take charge a part of an existing organization. The founder has own personal visions, goals, beliefs, values, and assumptions of how things should be which are initially imposed to the group. However, this does not automatically create culture but rather compliance to act as the leader asks for. If the resulting behavior leads to success, these founder's individual views become recognized and shared among the group which reinforces the cultural norms, and the group continues to behave accordingly. (Schein, 2017, pp. 130-132).

If the founder's beliefs and values do not lead to success, the group will fail and disappear or seek new leadership until a new leader with beliefs and values to success have found. The culture formation process of the group will then circulate around the new leader. With continued reinforcement, the group gradually internalizes these beliefs and values as non-negotiable assumption, and they come to be taken for granted. By this way they become part of the groups identity and are taught to newcomers how one should think, feel and behave in this group. (Schein, 2017, pp. 127-132). Thus, an organization culture is the result of a complex group learning process that is only partially influenced by leader behavior. It grows and develops over time and with new recruitments, whereby the new members of the organization adopt it as it is or possibly try to change it depending on their leadership role in that group, and the success history of the group.

Culture and leadership of an organization are tightly connected as leaders initially create a culture when they create an organization, and once established the culture starts to define how the organization defines the leadership it needs. Therefore, for leaders it is important to understand and work with a culture as they can influence the creation and development of it. Otherwise, if leaders are not aware of the culture they are involved in, those cultures will begin to manage them instead (Schein, 2017, p. 15, pp. 125-126).

2.1.4 Culture change mechanisms

A cultural change is one of the most challenging part of organizational development which is not possible to be implemented with top-down mandate as structural authority does not automatically create internal change movement but rather the opposite (Walker & Soule, 2017). According to Schein (2017, p. 7), once the organization has formed a set of shared basic assumptions, they can act as a defense mechanism to denial any potential change initiatives that could disrupt the achieve stability in that group. This resistance to change makes changing the basic assumptions challenging, time-consuming and highly anxiety-provoking. Therefore, the role of the leader in managing organization culture varies at the different stages of an organization's evolution as it grows and ages. This shift also impacts the processes and mechanisms by which culture evolves and changes, how effective they are, and how leaders can influence them. These change mechanisms, shown in Table 1, are cumulative which means additional ones become relevant as the culture evolves within the organization, while all previous ones are still present. (Schein, 2017, p. 233).

Table 1. Culture change mechanisms along organizational maturity stages (Schein, 2010).

Organizational maturity	**Change mechanism**
Founding and early growth	1. Incremental change through general and specific evolution 2. Insight 3. Promotion of hybrids within the culture
Midlife	4. Systematic promotion from selected subcultures 5. Technological seduction 6. Infusion of outsiders
Maturity and decline	7. Scandal and explosion of myths 8. Turnarounds 9. Mergers and acquisitions 10. Destruction and rebirth

Founding and early growth

In the founding and early growth stage, an organization's culture is shaped by its founders, forming the basis for member identity and guiding newcomers. Proposals to change the culture are often resisted unless an external crisis occurs, which might lead to leadership changes but typically preserves the core culture. (Schein, 2017, p. 234).

Incremental cultural change occurs through general and specific evolution, especially if the original founders remain involved. General evolution involves diversification, growing complexity, and higher levels of differentiation and integration. Specific evolution adapts parts of the organization to their environments, leading to cultural diversity within the core culture. This mechanism allows different industries to develop distinct cultures and subcultures within organizations, providing opportunities for significant culture changes later. (Schein, 2017, pp. 234-236).

Self-guided evolution through insight helps organizations assess and modify cultural assumptions for survival and effectiveness. Members collectively redefine their culture by changing core priorities or addressing barrier assumptions. Evolution can also occur by promoting insiders with new perspectives due to their personalities, experiences, or subcultures. These insiders have already adopted the core cultural elements, enabling gradual change without losing the organization's identity. Leaders must recognize what is missing in the current culture and select hybrids who best represent the new assumptions needed for the future. (Schein, 2017, pp. 236-237).

Midlife

Midlife organizations face pressures like adapting to new market conditions, financial challenges, or regulatory changes, forcing them to re-evaluate cultural assumptions and practices. Leadership transitions can be challenging, often leading to conflicts between conservatives favoring the founding culture and those seeking change. As the culture integrates into organizational structures, it becomes more resistant to change. Therefore, compatibility with existing cultural elements must be assessed during expansions, mergers, acquisitions, or new technological implementations. (Schein, 2017, pp. 237-239).

Leaders can evolve the culture by promoting individuals from advantageous subcultures into key roles. While this change mechanism is effective, it is slow and requires planned change projects to accelerate cultural transformation. Additionally, adopting new technology, known as technological seduction, can shift cultural assumptions. This can range from gradual technological diffusion to the managed introduction of specific technologies aimed at encouraging common behavior. However, unintended effects, such as increased hierarchical control, may occur. (Schein, 2017, pp. 240-243)

Infusing a new leader from outside often involves introducing their own team and removing those associated with old habits, disrupting the existing culture and initiating new cultural formation. However, this can cause more resistance compared to promoting insiders, and in case the new leader fails to improve the organization or threatens core cultural values, they may be discredited and replaced. Bringing in outsiders below top management can also stimulate culture change by reshaping subgroups and creating successful new models, influencing the broader organization when successful leaders from these divisions are promoted to senior positions. (Schein, 2017, pp. 243-245).

Maturity and decline

Continued success in an organization creates strong shared assumptions, forming a robust culture that is unlikely to be challenged. While it is beneficial in stable environments, this can become a liability in changing conditions. As organizations mature, they often develop a gap between espoused values and actual practices, leading to scandals when true operating assumptions are exposed. Additionally, members of the organization may leak information outside to expose such practices to the public to induce some change they want. While such scandals do not automatically cause culture change, they act as powerful forces that initiate change programs. (Schein, 2017, pp. 245-248).

If an organization is surprised by changing conditions, it may need rapid cultural transformation, often requiring strong transformational leaders. Turnarounds involve change programs that rebuild structures and processes, engaging all members to address dysfunctional elements and develop new assumptions. During mergers or acquisitions, cultural clashes are inevitable, with integration occurring either by one culture dominating or by blending elements from both. Reorganization and rebirth typically involve dismantling an existing organization to form a new one, which is often necessary for financial survival. (Schein, 2017, pp. 248-250).

2.2 Safety culture

Safety as a concept has many definitions depending on the context and scope of applications. The emphasis on safety varies depending on the industry as different industries face varying risks and hazards, leading to different understanding and approaches to safety. Additionally, in Finnish context, '*safety*' and '*security*' are often mixed or combined as the Finnish language only has one term '*turvallisuus*' to cover both of these aspects. The main difference between them is that '*safety*' is aimed at preventing accidents and other unintentional damages whereas '*security*' is aimed at preventing intentional acts which may cause harm to people, environment or property (IAEA, n.d.; Reiman & Oedewald, 2008, pp. 5-6). These both views on managing risks are needed to ensure the continuity of business operations which can also be referred as corporate safety and security (EK, 2016), or organizational safety and security depending on the context in question (Leppänen, 2006).

2.2.1 Definition of safety culture

Leadership and culture of an organization have been identified as essential aspects in safety and security management (EK, 2016), where '*safety culture'* is used to refer to how an organization's culture prioritizes and values organizational features that affect and influence safety of people, environment and property (Cooper, 2000, p. 113; IAEA, 1998, p. 3). The term '*safety culture*' was first introduced in 1980s by the International Nuclear Safety Advisory Group (INSAG), an expert group formed by the International Atomic Energy Agency (IAEA), in their report considering the 1986 Chernobyl accident (INSAG, 1991). The accident was concluded to be a result of several mistakes and violations in technical design and operation during the life cycle of the plant, where one of the major influencing factors was the poor level of safety culture on the organizational and national levels (INSAG, 1992). In their report, the INSAG (1991, p. 4) has defined safety culture as:

> *"Safety culture is that assembly of characteristics and attitudes in organizations and individuals which established that, as an overriding priority, nuclear plant safety issues receive the attention warranted by their significance."*

As can be recognized from the definition and its historical context, the INSAG (1991) gave particular attention on the organizations which are directly operating nuclear power plants as they have the closest link between human actions and nuclear safety, but they also recognized the possible contribution of governmental and other background and supportive organizations. This prioritization of nuclear safety is a unique aspect for nuclear industry, whereas in some other industries safety culture may refer for example to environmental, industrial process, or occupational safety. For a more general definition, regardless of the industry domain, Reiman and Oedewald (2009, p. 34) have defined safety culture as:

> *"Safety culture is the ability and willingness of the organization to understand safety, hazards and means of preventing them, as well as ability and willingness to act safely, prevent hazards from actualizing and promote safety."*

Additionally, many alternative views and approaches on safety culture and its relation to safety management exist and are used across various organizational environments. Some notable examples include the works of Cooper (2000) and Guldenmund (2000; 2010), among others. However, it is common that they share similar dimensions of leadership, questioning attitude, decision-making, accountability, learning, and communication. (Viitanen et al., 2022, p. 7). In response, the IAEA (2020) has been developing a harmonized safety culture model to provide guidance on overarching principles, aiming to align different safety culture frameworks, particularly for organizations operating in nuclear industry.

One commonly adopted theoretical framework to examine safety culture is based on Schein's (1985; 2017) three level organization culture theory, discussed in section 2.1.1. It is aimed to assess all levels of safety culture, as shown in Figure 2, considering the artifacts, values, and essentially the basic assumptions of the culture which guides the behavior in organization (IAEA, 2016c, p. 3).

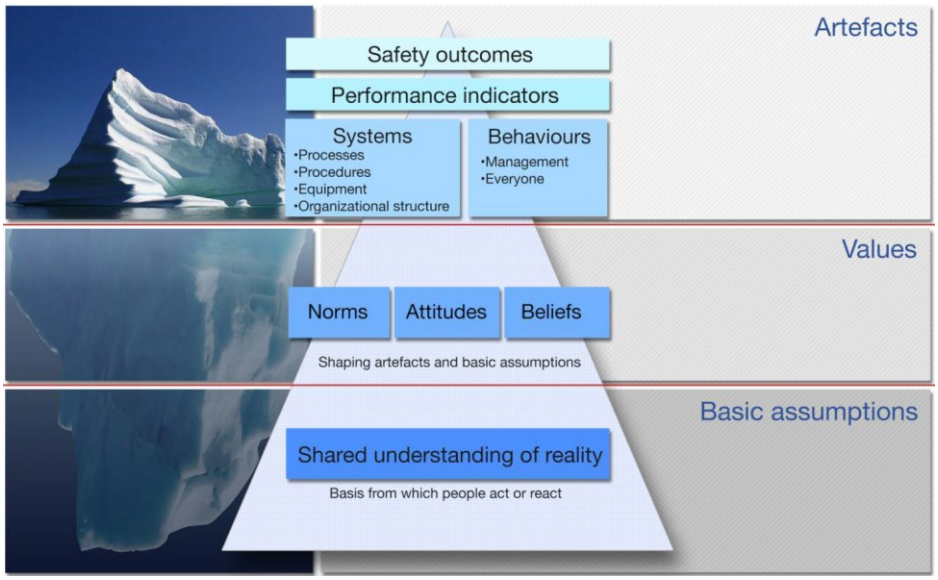


Figure 2. The iceberg model of safety culture (IAEA, 2016c, p. 4).

Safety culture is usually considered as a subset of an organization culture unless safety is the dominating factor of organization culture as in safety-critical industries (Cooper, 2000, p. 113; IAEA, 2002b, p. 3). Thus, there have been arguments that the concept of '*safety culture*' should be interpreted more as '*culture for safety*', as safety is not only some variable that can be added or removed from an organizational culture. Instead, safety should be approached with a holistic and systemic viewpoint, as illustrated in Figure 3. According to the IAEA (2016a), safety should be seen as an outcome of organizational culture, which varies in its capacity to understand and manage how their members behave, and how their management system is developed to support ensuring safety in everyday operation.

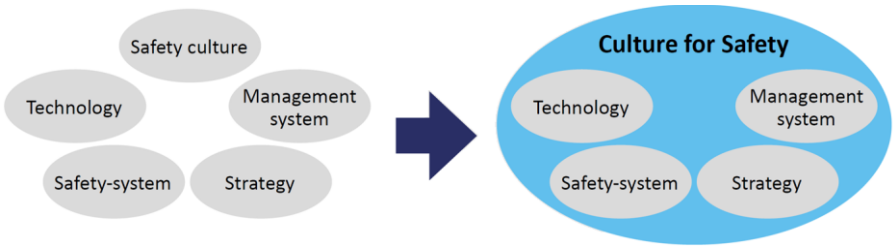


Figure 3. Compartmentalized and systemic approaches (IAEA, 2016a, modified).

Similarly to organizational culture, subcultures can emerge within an organization's safety culture by influence of different national or cultural backgrounds, or various occupations (Cooper, 2000, p. 113). This can lead to diverse views on safety and how it should be approached, with some preferring standardization and rules, while others adopt a more informal approach focusing on interpersonal relationships (Reiman et al., 2008, p. 21). These variable views and approaches to safety will influence behavior, potentially causing different outcomes depending on the compatibility of the subcultures.

Viitanen (2015, pp. 46-47) notes that when developing safety culture management strategy, understanding these subcultures is especially relevant in networked and project-based organizations where employee turnover is high, and a shared culture will not be able to form easily. Therefore, creating a safety culture that is shared by everyone may be desirable, but it may be practically not achievable with the reality of complex and networked organizations of today. However, Viitanen (2015) adds that a strong industry culture could help to create a common safety culture for otherwise loosely connected organizations.

2.2.2 Common indications of safety culture

While the systems and structures of an organization define and provide guidelines on how tasks should be accomplished, the reality is often very different when individuals need to adapt into certain situations with the skills, resources, and time they have available to achieve set goals. In some circumstances, these goals can be contradictory, achieving them might require bending the rules, or innovating outside the guidelines to get the work done. These small adjustments of procedures are not negligent, as they are usually driven by good intentions, but rather they may be indications of eroding safety culture if such manners become a common way to operate. (Reiman & Oedewald, 2009, p. 33; Reiman & Rollenhagen, 2018, p. 664).

The relevance of safety culture lies in organizations' tendency to drift into conditions where they struggle to identify vulnerabilities and the practices that create or maintain them. According to the IAEA (2016a), this can result from people gradually accepting declining conditions, prioritizing other concerns over safety, or ignoring risks because '*nothing has happened*'. Unlike accidents resulting from single malfunctions or errors, most incidents emerge from a systematic shift in work practices combined with unexpected events. These events often reveal that safety margins have eroded over time which may have remained undetected. (Reiman & Rollenhagen, 2018, p. 664). Investigations of such events in nuclear and other industries have consistently highlighted organizational and cultural root causes where identified common themes have included (IAEA, 2016c, p. 3):

- Insufficient understanding of operational reality by leaders
- Inadequate oversight and supervision of contractors
- Insufficient understanding of safety issues in decision making and actions
- Normalization of risks
- Failure to learn from previous events

Addressing such culturally related issues requires long-term and persistent efforts, and while they cannot be totally controlled, it is possible to influence them with safety related development activities. To help organizations assess and improve safety culture, the IAEA (2009, pp. 6-10; 2016c, p. 2) has developed a normative framework to provide guidance for assessing the cultural characteristics and attributes referred in their definition for safety culture. It consists of five high-level cultural characteristics which provide the basis for a strong safety culture:

1. Safety is a clearly recognized value
2. Leadership for safety is clear
3. Accountability for safety is clear
4. Safety is integrated into all activities
5. Safety is learning driven

Each of these high-level cultural characteristics are described more in detail with multiple attributes which act as international references to be considered when assessing and developing culture for safety (IAEA, 2016c, pp. 29-33; IAEA, 2020). Examples of such attributes, both positive and negative indications, are shown in Appendix A. In a similar approach, Oedewald (2015, pp. 15-18) has proposed that an organization has a high potential for safety when the following criteria are met:

1. Safety is a genuine value in the organization which reflects to decision making and daily activities
2. Safety is understood as a complex and systemic phenomenon
3. Hazards and core task requirements are thoroughly understood
4. Organization is mindful in its practices
5. Responsibility is taken for the safe functioning of the entire system
6. Activities are organized in a manageable way

A good safety culture does not mean that there should exist a unified agreement about everything in the organization. Diverse perspectives and opinions are important for improvement as an overly uniform culture can become blind to its own weaknesses which may reinforce those old and familiar habits. To promote a questioning attitude toward safety, the organizational climate should encourage open discussions about potential hazards, countermeasures, and their impact on business operations and public image. Simultaneously, an effective safety management system serves as both a sign of a healthy safety culture and a necessary precondition for its establishment within the organization. In this environment,

trust, openness, and accountability infuse the organization, empowering individuals to remain mindful about even minor deviations that could potentially escalate into significant safety issues. (IAEA, 2016a; Reiman & Rollenhagen, 2018, p. 658).

2.2.3 Organizational key factors for safety culture development

To be able to enhance the safety culture of an organization, one must first assess the current safety culture and then identify solutions to the detected problems, considering the organization's internal and external capabilities and circumstances. These solutions may vary across countries and organizations due to historical and cultural differences. According to the IAEA (2002a, pp. 8-9), there have been identified several organizational key factors which need to be considered when proposing solutions to recognized safety culture issues. The importance of each factor varies with the nature of the problem, and some factors have a wide-ranging impact on an organization's ability to effectively implement solutions.

Commitment from top management of the organization is a required ingredient to develop a strong safety culture, as they need to demonstrate their commitment to safety, ensuring it is not left solely to lower-level managers or safety experts. Actions to improve safety culture should be channeled through managers who visibly show interest in safety, motivating employees to become personally committed to safety enhancements. True commitment to safety means more than writing policy statements and giving speeches. It involves providing leadership and partnering with staff to translate safety goals into daily activities. This includes genuinely allocating time and resources to safety and ensuring managers are trained and competent in safety related matters. (IAEA, 2002a, pp. 9-10; INSAG, 2002, p. 5).

Continuous improvement and learning have been identified being important for any organization, regardless of its current performance, as they need to consistently search for improvements in its systems and processes. A continuous improvement attitude must be ingrained within the organizational culture before attempting to develop a robust safety culture. Employees at all levels must be involved in the learning and improvement process, contributing ideas for improvement and recognizing the benefits of these enhancements. Without a continuous improvement attitude, organizations risk falling into self-satisfaction and organizational inertia, where performance is superficially checked against targets without genuine attention to the organization's overall health, employee capabilities, and the quality of its culture. (IAEA, 2002a, p. 9; INSAG, 2002, pp. 10-11).

Communication is essential for any organizational activities and especially for developing and improving safety culture as organizations must ensure information is effectively communicated internally and to external stakeholders and organizations. Good communication channels facilitate the sharing of knowledge and experience among organizational groups, encouraging learning and collaboration. Information transmitted within the organization can be susceptible to ambiguity and distortion, leading to conflicts and hindering collaboration. This makes it necessary to establish formalized communication channels and processes to ensure consistent and effective information flow. Communication with external organizations should follow the same principles to ensure transparency and trust between parties. (IAEA, 2002a, p. 9; INSAG, 2002, pp. 11-13).

Additional organizational factors identified by the IAEA (2002a, pp. 7-10) and the INSAG (2002, p. 5) include conservative decision-making, fostering a challenging and questioning attitude towards concerning acts or conditions, and promoting a blame-free reporting culture. Therefore, organizations should encourage their members to thoroughly understand situations when making decisions that could have safety-related outcomes, confidently challenge potentially unsafe practices, identify deficiencies whenever they encounter them, and report even minor near-misses for organizational learning purposes. These organizational factors are still recognized as key safety culture traits by the IAEA (2020) in their latest working document towards a harmonized safety culture model.

2.2.4 Development stages of safety culture

Organizations operating in safety-critical environments share a common concern for safety, but they may often have a different understanding of safety culture. This variation may reflect different levels of awareness regarding the safety impact of human behavior and attitudes. Over time, when organizations strengthen their safety culture, they typically go through multiple phases which could be represented as three main stages to simplify the complex process of cultural development (IAEA, 1998, pp. 5-7; INSAG, 2002, p. 3):

I. Safety is compliance driven and based on rules and regulations
II. Safety is considered as an organizational goal
III. Safety is seen as a continuing process of improvement

As the cultural development is a complex process of organizational learning, it is possible that organization may have any combination of the characteristics of these stages, and therefore they act more as guidelines for safety culture development (IAEA, 1998, p. 5). However, it is important to recognize that achieving the cultural characteristics of the last

stage might not be possible without first going through earlier stages of compliance culture and engineered management systems which needs to be maintained while progressing towards issues related into more human aspects (INSAG, 2002, p. 4).

In the first stage, an organization views safety primarily as a technical issue, aiming to meet external requirements rather than integrating safety into its core values. Compliance with rules and regulations is considered sufficient, and the organization has limited awareness of how behavior and attitudes impact safety performance, showing no willingness to address these issues. Such organizations tend to solve problems reactively and blame people when mistakes occur. Collaboration and communication between departments are poor, sometimes leading to internal competition. Employees are defined by their roles, often believing that safety is management's responsibility, rather than actively participating in it. Management, in turn, focuses on endorsing rules, pushing employees, and expecting results, often prioritizing short-term profits over long-term safety and stability. (IAEA, 1998, pp. 5-6; INSAG, 2002, p. 3).

In the second stage of development, the organization understands the importance of safety performance without external requirements. It develops safety-related vision, values, and goals, and establishes systematically documented processes and procedures to achieve them, though these remain on the technical side and do not include behavioral aspects. These improvements raise safety awareness but do not increase individual and team commitment, as safety is still often determined by externally set rules. In this stage, day-to-day matters take priority over strategic planning, focusing on operational efficiency at the expense of long-term vision. Management promotes cross-departmental collaboration to enhance communication and knowledge sharing, discouraging conflict in favor of teamwork. Decisions often revolve around cost, with safety, cost, and productivity sometimes seen as conflicting priorities. Short-term profit goals take precedence, even rewarding individuals for exceeding targets regardless of long-term consequences. There is growing awareness of cultural issues, but it is not understood why additional controls do not result in expected safety performance. (IAEA, 1998, pp. 6-7; INSAG, 2002, pp. 3-4).

The third and final stage means that safety has become part of daily operations with a fully shared safety-related vision and values that most individuals are committed to. The organization has adopted continuous improvement where everyone can contribute to improve organizational efficiency and effectiveness. Insufficient practices are openly challenged, and incidents are seen as avoidable exceptions whether they are safety related or not. There is a high level of awareness of how behavior and attitudes influence safety, and

measures are taken to improve behavior, leading to a learning organization with a self-sustaining safety culture. In this stage, the organization acts strategically, focusing on long-term performance to proactively address problems. Collaboration between departments is supported, and conflicts are resolved through mutually beneficial solutions. Mistakes are viewed as opportunities for learning and improvement, rather than occasions for blame. Safety is integrated into decision-making and does not conflict with productivity or profitability. Management's role evolves into coaching, with people respected and valued for their contributions, whether they produce results or support others. (IAEA, 1998, p. 7; INSAG, 2002, p. 4).

2.3 Organizational development interventions

In the dynamic business environment organizations need to continuously adapt and go through organizational development interventions to remain competitive. This demand for organizational change is closely tied to business strategy and the operational environment, including internal and external factors of disruption. This study utilizes organization development and change intervention practices presented by Cummings et al. (2019) which is a highly recognized course book in the field of organizational development as it offers a comprehensive and well-structured framework for understanding and implementing development interventions.

According to Cummings et al. (2019, pp. 141-145) an organization development intervention involves a sequence of activities, actions, and events aimed at enhancing an organization’s performance and effectiveness. They aim to address specific problems and improve specific areas of organizational functioning identified during the development process. To achieve the intended outcomes, these interventions must be planned to fit the organization’s needs and consider various aspects of its members, as well as the challenges arising from the change process itself.

2.3.1 Planned change process

Planned change makes a distinction to changes that happen to an organization from changes that are planned by its members. Organizations can use planned change to solve their organizational issues, adapt to external changes, learn from experiences, and improve their performance. All approaches to organizational development are based on some theories of planned change, which describe the stages and processes of how change can be implemented and managed in the organizations. (Cumming et al., 2019, pp. 20-21).

From historical perspective, one of the earliest models of planned change was developed by Kurt Lewin (1947), which considers organizational behavior as a result of two opposing forces: those maintaining the status quo and those pushing for change. When these forces are about equal, the organization's behavior is stable. For one to be able to change the state of an organization, this harmony needs to be disturbed by either increasing forces pushing for change or decreasing forces maintaining the current state, or some combination of both. (Cummings et al., 2019, pp. 21-22; Schein, 2017, pp. 322-323). Lewin's model for planned change process consists of three steps:

1. Unfreezing: reduce forces that maintain current behaviors.
2. Movement: shift behavior with new values, attitudes, structures and processes.
3. Refreezing: stabilize new equilibrium with culture, rewards and structures.

This simple model is widely cited, also by the IAEA (2002b, pp. 60-64), and it has laid a foundation for understanding of organizational change, which has been elaborated further and more detailed by many other models of planned change (Cummings et al., 2019; Williams, 2017). However, organizations tend to be more complex than this simple model describes, and it is unlikely that change ever truly end due to the continuous external and internal pressures that modern organizations need to manage. Generally, more recent models of planned change still involve similar key activities, as shown in Figure 4. These activities are interconnected, highlighting that organizational change is not linear or a one-shot event, but rather an iterative process (Cummings et al., 2019, p. 27).

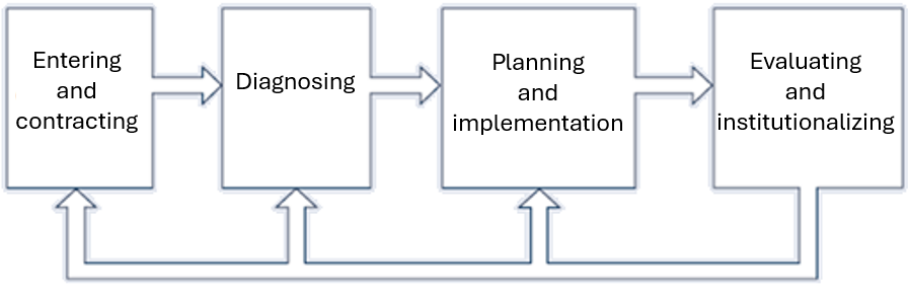


Figure 4. General model of planned change (Cummings et al., 2019, p. 28, modified).

Entering and contracting are the first steps in planned change which helps leaders to decide whether to proceed with a change program and commit resources into it. This phase involves gathering initial data to understand organizational problems or opportunities. The collected information is then discussed with other members of the organization to develop a contract outlining future change activities and resources. Many organizations do not progress beyond this phase due to disagreements, resource constraints, or as more feasible alternatives have been found (Cummings et al., 2019, p. 28).

In the diagnosing phase the organizational system is thoroughly examined to understand the causes and consequences of its organizational issues. This activity involves selecting a suitable diagnosing model, gathering, analyzing, and feeding back information to members about existing issues or opportunities. Diagnostic models can explore activities in three levels including the organization, group, and individual levels. Data for this analysis is collected through interviews, observations, surveys, and going through documented information, which are then analyzed and reviewed. (Cummings et al., 2019, p. 28).

In the planning and implementing phase, organization members collaboratively design and execute organizational development interventions to achieve their desired goals. Intervention design is based on criteria like readiness for change, existing capabilities, culture, and power distribution in the organization. There are four major types of interventions which are tailored according to the findings of the diagnostic phase (Cummings et al., 2019, pp. 28-29):

- Human process interventions (e.g. communication, decision making, leadership)
- Technostructural interventions (e.g. job design, hierarchy, employee involvement)
- Human resource interventions (e.g. performance and talent management, rewards)
- Strategic interventions (e.g. external environment, organization design, culture)

The final phase involves evaluating the outcome and effects of the intervention, and institutionalizing changes to make them persist in the organization. This includes providing feedback to organization members about the results to decide whether to continue, modify, or suspend the changes. Finally, institutionalizing successful changes involves reinforcing them through feedback, rewards, and training. (Cummings et al., 2019, p. 29).

2.3.2 Transformational change

Transformational change is a type of planned change that involves making strategic choices to fundamentally change organizational culture, processes and structures to adapt to new conditions and to support the new strategic direction. It is typically triggered by significant external or internal disruptions that threaten organization's survival and requires significant changes to its business strategy and organizational elements. Studies show that organizations experience periods of smooth growth and incremental changes which are interrupted by severe disruptions. Successful organizations respond by transforming to fit new conditions rapidly which allows them to quickly benefit from new ways of operating, gaining a competitive advantage over slower-changing organizations. This response helps organizations reach another period of smooth operation until the next disruption triggers a

need for a major change. Because of the major impact for the target organization, a transformational change is typically driven by senior executives of the organization who decide when to initiate it, what should be changed, and how it should be done. The role for the transformational leader is to clearly communicate the new direction and ways to achieve it, provide energy to fuel up the change, and enable the change by providing the necessary resources for it. (Cummings et al., 2019, pp. 446-449).

2.3.3 Continuous change

Continuous change interventions extend transformational change into an ongoing process of learning, strategy setting, organization designing, and implementing change. Learning organizations continuously evolve by empowering their members to take responsibility for change and learning which involves two key processes: organizational learning and knowledge management. Organizational learning enhances the ability to acquire and develop new knowledge by focusing on organizational structures and social processes that enable members to learn and share knowledge. Knowledge management on the other hand focuses on organizational tools and techniques for collecting, organizing, and translating information into useful knowledge to be used to improve its performance. Together, these processes lead to enhanced capabilities of continuous improvement for the organization. (Cummings et al., 2019, pp. 478-479, pp. 490-491).

In contrast to individual learning, learning is considered organizational if it achieves organizational purposes, is shared among members, and is embedded in its systems, structures, and culture. Thus, individual learning may not always contribute to organizational learning and vice versa. For example, individual members may acquire new skills which are not integrated into the operations of the organizations, even though they might be utilized to achieve the goals of the organization. Alternatively, improvements in organizational processes reflect organizational learning even if they are not understood by individual members. Thus, one of the main characteristics for organizational learning is that it remains within the organization even if its members change. (Cummings et al., 2019, pp. 491-492).

Self-designing organizations can fundamentally and continuously adapt themselves to changing conditions in turbulent environments through a highly participative process that empowers constant innovation and learning. It contrasts with traditional change methods which rely on periodic events with pre-made plans. The dynamic nature of self-designing approach demands continuous organizational learning, where its members iteratively innovate by trying new structures, processes, and behaviors. This approach emphasizes

considering diverse goals and interests to gain stakeholder support and equipping them with necessary knowledge and skills, with top executives formulating strategy and vision, which is then operationalized at middle and lower levels through structures, procedures, and behaviors. (Cummings et al., 2019, p. 479, pp. 486-489).

Built-to-change approach designs flat and flexible organizations to be responsive to continuous change, unlike traditional organizations that prioritize stability and avoid changes which may hinder their adaptability in rapidly changing environments. Built-to-change organizations focus on selecting and developing talent that thrives on change, and they also implement reward systems that are tied directly with change-related goals which motivates to continuously develop new skills. Transparent and up-to-date information allows organizations to stay competitive by making timely and relevant decisions. Shared leadership is one key aspect of built-to-change organizations, with responsibilities distributed across multiple levels to speed up decision-making and response times. This approach develops leadership skills across a broader range of members, creating a stronger leadership basis throughout the organization. (Cummings et al., 2019, p. 479, pp. 500-501).

2.3.4 Transorganizational change

Transorganizational change involves interventions beyond a single organization, becoming common as organizations merge or acquire others, form alliances, or collaborate on shared business operations. This approach requires coordinated strategies, goals, structures, and processes across organizations, which increases complexity and potential for conflicts. It addresses unique and often conflicting needs by linking organizations to share risks, resources, and expertise for large-scale projects that are too costly and complex for single entities. (Cummings et al., 2019, pp. 509-511).

Strategic alliance interventions, such as joint ventures or long-term partnerships contracts, help organizations develop collaborative relationships that offer mutual benefits despite reduced autonomy. These arrangements require a clear understanding of goals, trust-building, and ensuring expected benefits. A strategic alliance is a formal agreement between two or more organizations to pursue shared goals by sharing resources like intellectual property, people, capital, technology, and physical assets. Effective strategic alliances follow a process of strategy formulation, partner selection, alliance structuring and start-up, and ongoing operation and adjustment. These interventions enable organizations to

leverage each other's strengths for competitive advantage, facilitating resource and capability sharing. (Cummings et al., 2019, pp. 518-519).

Network interventions enable groups of organizations to collaborate on complex tasks that single entities cannot handle alone. These loosely coupled, nonhierarchical systems help organizations recognize the need for partnerships and develop coordinating structures. They are increasingly popular as they allow organizations to share resources and risks particularly in large-scale efforts. Managing these networks involves two types of change: creating the initial network and managing change within an established network. Initial creation focuses on understanding relationships and roles among participating organizations, while managing change within networks requires viewing the relationships as a whole system. Networks produce complex phenomena that cannot be fully explained by their individual parts as each organization has goals related to both the network's good and its own self-interest. Therefore, change in complex systems involves creating instability, managing the tipping point, and relying on self-organization. However, understanding the whole system and interdependencies is essential, as individual actions based on self-interest can lead to systemic issues. (Cummings et al., 2019, pp. 522-527).

2.4 Summary and discussion of literature review

Organizational culture emerges and develops under the influence of its leaders, and it has a significant impact on the members of the organization as well as its operations through behaviors, attitudes, and values. This culture guides members in justifying operations and making decisions, especially in conflicting situations where individuals may seek support on how to proceed. Culture creates a tight bond between an organization and its members which makes it difficult and time-consuming to change. (Schein, 2017; Williams, 2017).

When considering safety culture in organizations (IAEA, 2020; Reiman & Oedewald, 2009), it should be interpreted as the organization's ability and willingness to understand and prioritize both direct and indirect operations that could have immediate or delayed safety-related outcomes. This includes actions and decisions made or not made, meaning both active and passive operations should be considered. According to Weick (1998, p. 74), organizations and their activities are strongly defined by what they choose to ignore and how widely spread and consistent this is within the organization. Thus, safety should approach with a holistic and systematic approach where the overall performance of the organization will influence safety outcomes and vice versa (IAEA, 2016a; Oedewald, 2015). If illustrated with an exaggerated example, safety either is or is not ingrained into the operations of the

organization. Management of the organization has a major role and responsibility in creating, changing and sustaining organization culture (Schein, 2017; Cummings et al., 2019), and how it prioritizes safety with other business objectives (IAEA, 2016c; Milch & Lauman, 2016; Oedewald, 2015). Based on the literature review and observed industrial behavior, following hypotheses are formed to guide the preparation of interview questions:

- Open and transparent communication between organizations enhances collaboration efforts and contributes to the development of a safety culture.
- Joint initiatives and shared safety goals between organizations lead to a more unified safety behavior and attitudes.
- Top management's commitment to develop safety culture often conflicts with other business objectives such as profitability.
- Business solutions, such as collaborative business models or financial intensives, can be used to significantly influence the development of safety culture.

These hypotheses presume that open and transparent communication, collaboration, and joint initiatives between organizations are fundamental to enhance the development of a unified safety culture. These can be enforced with collaborative contractual formats such as forming an alliance or some other collaboration model than traditional contractual arrangements. Additionally, the challenging role of the top management is considered to include decision-making conflicts between business operations and financial of the organization as is anticipated by the executive subculture they usually represent (Schein, 2017, pp. 226-228; Reiman, 2009, p. 15).

3 Methodology

This chapter covers the research methodology of this study. First, the research approach is explained and justified. Next, the data collection method and interview details are outlined. This is followed by an overview of the thematic data analysis used in this study. The final section provides an evaluation of the research process and discusses the ethical considerations of this study.

3.1 Research approach

Research methods related to safety culture can be broadly divided into two categories based on whether they are quantitative or qualitative. Quantitative methods provide large amounts of data to be analyzed with questions often related to measuring how good something is on a specific scale. Examples of such data collection methods include surveys and checklists related to audits. On the other hand, qualitative methods include interviews, group discussions, and observation, which provide more in-depth information about the research topic to describe how something is like or why it exists in its current form. (Reiman et al., 2008, p. 27).

To address the research questions, a qualitative approach is chosen over a quantitative approach due to its suitability for deeper exploration of subjective perspectives, experiences and context-dependent nature of safety culture in safety-critical environments. According to Eriksson & Kovalainen (2008, pp. 5-6), qualitative research is particularly relevant when more subjective and interpretative research is needed to address unstructured problems which are often related into phenomena with modest insights or have remained unclear in quantitative studies.

This research uses semi-structured, open-ended interviews as a data collection method, which proceeds like conversation around predefined themes set by the interviewer. Semi-structured interviews were chosen due to the broad nature of the research topic related to safety culture. This approach allowed deepening and refocusing of interview questions with tailored follow-up questions as needed, giving interviewees the opportunity to discuss issues more broadly within the defined themes (Eriksson & Kovalainen, 2008). Additionally, in safety culture studies interviews have been found to provide more detailed and critical information about visibility of leadership and organization learning than what can be gathered with surveys (Reiman et al., 2008, p. 33). However, compared to surveys, interviews require more time per participant, which is why they are usually conducted with

a smaller number of individuals. This is to keep the acquired data highly qualitative even though limitations in quantity, and limit the time required to set up the interviews, transcribe and analyze them. (Eriksson & Kovalainen, 2008, pp. 33-35).

3.2 Data collection

Target group for the interviews considered managers and leaders with experience on safety-critical industries in Finland. Participants for the interviews were approached via email utilizing contacts of personal professional network created during my working career. This included connections working in organizations operating in nuclear, hydrogen, energy production, petrochemical, and chemical industries either as business operators, technology vendors, or engineering and project service providers. It was considered valuable if the interviewees would have experience from multiple industries or organizations within them, as this would enable them to reflect on and compare their experiences in these different environments.

Interview questions were developed based on the literature review and formulated research hypotheses, integrating insights from personal experiences related to identified safety culture challenges. Interviews comprised of ten predefined open-ended questions to guide the interviews to cover at least certain interesting themes according to the research hypotheses, but also to leave a possibility to ask clarifying questions based on the interviewee's responses to gather additional information in case some valuable insights emerged during these interviews (Eriksson & Kovalainen, 2008, pp. 80-82; Reiman et al., 2008, p. 33). Main themes of interview questions covered following topics:

- Appearance and recognition of safety culture on different industries
- Influencing behavior and attitudes in internal and external organizations
- Enabling effective collaboration over organizational boundaries
- Blaming culture and countermeasures to reduce it
- Safety considerations and conflicts during decision-making process
- Relation between safety and business aspects of an organization

The empirical data for the study was gathered with six individual online video interviews during September and October of 2024. They were held in Finnish based on interviewee's preference and lasted approximately 60 minutes According to Terry et al. (2017), the recommended sample size for this type of thematic study with interviews is typically between six and fifteen. Initially, the intention was to interview up to ten

individuals, but it was soon recognized that answers started to repeat after a few interviews. Therefore, the remaining unscheduled interviews were not conducted. It was also common for upcoming questions to be partly covered, but in these cases, additional follow-up questions were raised to deepen the discussion.

The professions of the interviewees ranged from team leaders, project managers and business managers to vice presidents and executives of the organization, covering operational, technology, and business management functions of the organizations. The organizations interviewees currently represented varied in their maturity level, from established corporations to growth companies. All of the interviewees had experience in multiple industries or organizations, which provided valuable insights. In addition, most of the interviewees had experience from international business environments which offered further insights into the influence of national cultures. These factors enabled a wide range of industrial perspectives despite the limited number of interviewees.

3.3 Data analysis

After collection the empirical data through interviews, the data analysis part was conducted using a thematic analysis approach. This widely recognized method involves the systematic identification, analysis, and reporting of patterns, or themes, within the collected data (Braun & Clarke, 2006; Guest et al., 2012). This approach aligns well with the intention to identify common factors and themes from the interview data.

The analysis process began with transcribing the interviews using the automatic transcription function of Microsoft Teams. These transcriptions were verified by cross-checking them with the original audio recordings. In two instances, the transcription was done manually based on the audio recordings, as the automatic transcription was not properly functional. Following the transcription of all interviews, the data was analyzed by categorizing and combining it according to certain themes.

The approach to generating codes combined deductive and inductive methods, meaning that codes were identified both from theory and the data. Deduction is a linear process for building theoretical knowledge, starting with theory to deduce hypotheses, which are then empirically tested (Braun & Clarke, 2006; Eriksson & Kovalainen, 2008, p. 22, p. 129). The inductive method was used to recognize additional themes from the data not linked to the prepared hypotheses and guiding interview questions. After this phase, the extracted data was organized under different themes that corresponded to the specific research questions. According to Guest et al. (2012), segmenting text, coding, and

identifying themes are interconnected processes that occur simultaneously. In this thesis, the data was segmented and coded based on the guiding hypotheses and interview questions. Structural coding was guided by a predefined framework, and the codebook was developed based on the interview structure to ensure consistency and reliability in the data analysis.

Furthermore, an additional layer of data analysis was applied to the interview data using Schein's (2017) three-level culture framework discussed in section 2.1, which has also been adopted by the IAEA (2002a; 2002b; 2016c) to analyze elements of safety culture as discussed in section 2.2. This framework helps to explore cultural aspects by examining artifacts, espoused values, and basic underlying assumptions. By applying this framework, the analysis not only identifies surface-level observations but also tries to understand the deeper elements that influence organizational behavior and attitudes. This comprehensive approach ensures that all levels of safety culture are considered, providing a more thorough understanding of the safety culture considerations within the safety-critical industries.

3.4 Evaluation of the research process and ethical considerations

The reliability and validity of the research can be assessed by determining whether its conclusions accurately describe or explain the research questions and are supported by evidence (Eriksson & Kovalainen, 2008). Given that the research is based on a small number of interviews, it provides a narrow view of the questions being studied. Furthermore, its reliability heavily depends on the informativeness of these interviews, which in turn relies on how they are targeted and the selection of interviewees (Reiman et al., 2008, pp. 32-33).

Regarding the reliability of the research, this study assumes that the interviewees have no interest in being misleading which means that the similar data could be collected again. In terms of validity, this thesis uses triangulation by comparing the findings of the interviews with each other and with theoretical frameworks from the literature. However, the results of this of study should not be seen as a final truth as they are always subject to interpreter biases, especially when there does not exist a diverse team conducting this study to challenge others to minimize the risk of biases (Guest et al, 2012; IAEA, 2016c, p. 26). Furthermore, all interviewees and the author have their own biases, assumptions, and expectations, which may affect the findings of this thesis.

According to Eriksson & Kovalainen (2008), key ethical considerations in business research include informed consent, confidentiality, anonymity, and professional integrity. Participation in the interviews for this thesis was voluntary, and participants were informed about the purpose of the research. Interviews could have been interrupted or ended at any

time, and questions could have been skipped if they felt so. Confidentiality and anonymity were ensured and remained a priority throughout the research process. Additionally, it is important to note that no sensitive or confidential information was used in this research, and the participants were representing only themselves as a person instead of their current organizations. However, most of them mentioned that the values and perceived cultures of their organization definitely have influenced their personal views.

Furthermore, one ethical consideration about the interviewee selection is that they were not randomly selected, and all of them were personally known by the author before this research was conducted. This affiliation may have granted an easier access to the interviewees, but it also may have influenced who got invited into this research. This could predispose to inviting individuals with favorable opinions and attitudes instead of only belonging to the target group of the research.

4 Findings

This chapter presents the main findings of the empirical data gathered through semi-structured interviews. The interview data was analyzed using themes identified from both the literature and the data itself. These themes have been organized according to the research questions, starting with the appearance and recognition of safety culture, followed by influencing practices and the factors contributing to the development of safety culture.

The data is described as discussed during the interviews, with key takeaways cited as direct quotes. The themes of the findings are presented in the order that the topics emerged during the deepening discussions in the interviews. A more in-depth comparison of the findings with the reviewed literature is discussed in section 5.1, which also draws conclusions from this analysis. To conclude this chapter, the findings from the applied organization culture and safety culture frameworks are presented.

4.1 Appearance and recognition of safety culture

4.1.1 Strategical basis for business operations

At the beginning of the interviews, the discussion focused on the interviewees' general views on what safety culture means to them and how it appears within organizations or their businesses. From the statements collected in the interviews, it is clear that safety culture is an essential part of the foundation of an organization's business, regardless of the industry. The central role of safety culture is clearly evident in strategic priorities, and its effects extend widely across various aspects of operations.

> *"Safety culture is part of our DNA, meaning safety is present everywhere, starting from strategic priorities."* (Interviewee, translated)

Thus, safety culture is not just an individual practice but is deeply integrated into the core of the organization, guiding people's behavior and being continuously present in all activities. Interviewees described that the presence of safety in all operations creates a strong foundation upon which profitable business can be built. Without safety, there can be no profitability, and maintaining a good safety culture directly reflects the success of the organization. Therefore, safety and profitability are not in conflict but complement each other: safety culture creates the foundation upon which profitability is built, making it a key pillar of successful business operations.

"If you manage safety culture well, profitability and other business benefits will follow." (Interviewee, translated)

Safety culture should be embedded throughout the organization so that it is part of its processes from the beginning. This ensures that the organization's actions and omissions do not lead to unsafe acts in, for example, the processes of energy and production plants, construction sites, for the end-users of services or products, or the surrounding environment and property. The presence of safety culture at all levels of the organization is crucial, and it should be reflected in long-term goal setting and operational planning. For instance, all potential safety aspects should be considered in the early phase evaluations of investment projects and product development to ensure safety throughout the lifecycle of them, including during the development, design and execution phases as well as over the long term. This is particularly important during the construction phase and for the coming decades as facilities are used, renewed, or improved. A good safety culture appears in practical ways, such as avoiding carelessness, which helps prevent unfortunate incidents like accidents and injuries, thereby reducing extra costs and improving the quality of operations.

"Safety culture must be present at all levels of the organization so that individual project organizations do not have to separately instruct and define how to enable safe practices in their activities and delivered products." (Interviewee, translated)

In the nuclear industry, safety is a particularly significant value, and it is emphasized much more in the organization's operations than in other industries. Nuclear safety is a unique feature of the industry, addressed at all levels of the organization. Safety issues are always present in corporate management meetings, and in the context of nuclear power, safety needs to be seen as a '*license to operate*'. However, people working in the nuclear sector may become blind to how well things actually are when compared to many other industries. Interviewees from other industries also shared similar thoughts and views on safety culture which highlights the importance of safety and its central role in the organization's success.

4.1.2 Indicators, metrics and assessment

As part of the interviews, discussions focused on the observation, measurement, and assessment of safety culture and related phenomena in various environments and contexts. Interviewees generally indicated that safety culture, or its easily observable outcomes, can

be illustrated with various references or measured values, but determining its overall level can be challenging. Safety-related measurement is based on risk-oriented thinking, emphasizing the identification, reporting, and prevention of risks. Both reactive and proactive indicators were discussed during the interviews.

> *"Some things like work accidents can be easy to measure, but generally, the level of safety culture is difficult to measure."* (Interviewee, translated)

Reactive indicators, such as monitoring work accidents and near-misses, provide insights into the organization's safety status but focus on past issues and do not directly help prevent future hazards. For instance, work accidents relative to man-hours worked are commonly measured across various industries. Similarly, production plants can generally measure emissions and incidents, or in nuclear safety, the number of scrams (rapid emergency shutdowns) or other operational disturbances.

However, the problem with reactive indicators is that they focus on the past. According to the interviewees, a better approach would be to use more proactive indicators that help identify and prevent problems before they occur. Proactive indicators include observation rounds, safety rounds, and observation reports, which can be conducted daily and involve all organizational levels, from management to employees.

> *"You cannot just look in the rearview mirror; you need more forward-looking indicators."* (Interviewee, translated)

Proactive indicators aim to measure how actively an organization identifies and responds to safety risks. They promote openness and learning within the organization, allowing changes in operations, communication, and training before hazards occur. In developing and implementing proactive indicators the top management’s commitment to safety culture is crucial. When top management is genuinely involved in safety work and allocates time and resources to it, this commitment is reflected throughout the organization. For example, safety rounds and observations conducted by the top management can serve as indicators of their commitment and proactive approach to promoting safety.

Assessing the level of safety culture and verifying the accuracy of presented claims are central elements of organizational safety management. Based on the interviewees' views, it is evident that although it can be challenging to demonstrate the level of safety culture in advance, it can be assessed in various ways, such as through surveys, inspections, and audits.

"The level of safety culture can be hard to demonstrate or assess in advance in a way that it really matters." (Interviewee, translated)

Conducting surveys systematically collects information about the organization's safety culture and allows for benchmarking against other organizations. Safety culture surveys can reveal employees' views on safety matters and indicate how well safety culture has taken root in the organization which provides valuable comparative data and helps identify potential areas for improvement in operations. Audits and inspections are also effective tools for assessing safety culture, as both internal and external audits can evaluate how well various safety aspects are considered in organizational operations. These processes examine documentation and actual practices, helping to reveal any discrepancies between claims and reality. For example, based on the interviewees' practical experience, organizations' quality certificates and other documents may look good, but daily operations may not always align with what is described in the documents. Therefore, audits and inspections can thus uncover hidden issues that might otherwise go unnoticed.

"In these cases, papers can be made to look exactly as desired." (Interviewee, translated)

It is desirable and also possible to pre-assess the level of safety culture before starting collaboration between organizations. For example, technology vendors and service providers can be asked to provide their safety procedures and plans already at the inquiry stage or even earlier when potential partners are being identified. This procedure aims to ensure that safety views and practices are in-line before the chosen partner begins execution. Additionally, requesting comprehensive documentation in advance can reveal any shortcomings in the organization's readiness and ability to act according to its promises, making it easier to address and implement corrective or supplementary measures if seen so.

4.2 Practices to influence elements of safety culture

4.2.1 Unified understanding of shared goals and ambitions

In discussing ways to influence inter-organizational collaboration and particularly the development of the safety culture within it, the significance of forming common goals and shared will frequently emerge during the interviews. Safety culture is always rooted in people and their behaviors, and it can be developed in a desired direction by setting clear goals and clarifying the desired behavior patterns, such as the organization's values that guide towards a good safety culture. For inter-organizational collaboration to be effective, it

is important that organizations have common goals and time to interact with each other. Creating strategies and business plans together and spending time together lay the foundation for effective collaboration. This approach helps create a common understanding and set clear expectations, which is also important for ensuring safety.

> *"Potential business conflicts of interest can lead to situations where each party pursues their own interests. We should strive to have more of a 'one hat' approach."* (Interviewee, translated)

Although partner organizations may not always know how to address safety-related issues, it is important to clarify them together. Additionally, within a single organization, there may be several separate business units, leading to situations where each one seeks to advance only their own interests. A similar conflict of interest can arise in inter-organizational collaboration. In such cases, a culture that encourages collaboration and shared goals and will is more decisive than legal resolutions. The goal of collaboration is to create a common understanding and set clear expectations, reflecting a growing trend towards more open and collaborative operations.

> *"Address issues as they arise, rather than brooding over them."* (Interviewee, translated)

At the beginning of collaboration, it is important to agree that matters are handled in collaboration and in the spirit of partnership. Especially, when working on new kinds of business matters, one should not immediately resort to breaches of contract but raise issues as they arise and discuss them together before escalation. The focus should be on finding solutions rather than blaming, and issues should be addressed immediately as they arise to avoid potential conflicts.

> *"Something may have been sold differently than bought. The content of what we were supposed to do together may be perceived differently by the parties. There is a different understanding of what we were supposed to do together."* (Interviewee, translated)

Quality and safety-related expectations can be set for external partner organizations and included in collaboration agreements. If these matters are not agreed upon in the contract, it is important to agree on the goals and methods of collaboration as soon as possible. This means not only examining what has been agreed upon but also identifying what has been left unagreed and addressing those.

4.2.2 Commonly agreed ways to collaborate and communicate

Establishing mutually agreed upon operating procedures and communication methods is crucial for creating a safety culture in inter-organizational collaboration. Organizations involved in the collaboration should agree on how to collaborate and communicate, and how to adhere to and monitor these methods as early as possible. Clear, common rules and standards enable the formation of more compatible business operations and support consistent practices that promote a unified understanding of safety related topics.

At the beginning of the collaboration, it is beneficial to spend time agreeing on common goals and procedures. For instance, during project initiation time can be allocated for the parties to get to know each other better and strive to create a collaborative culture. The aim is to ensure that all parties understand what the collaboration is about, what is being done, and how it should be done. Collaboration methods should be well-managed and traceable so that new participants can easily understand the principles and goals of the collaboration.

Open and transparent communication is essential in organizational collaboration as information must flow freely and efficiently between organizations so that all parties are aware of the situation and potential risks. Open communication and feedback, both positive and negative, ensure that information circulates within the network as effectively as possible. For example, meetings can start with EHSQ (environment, health, safety, quality) issues to keep safety related matters continuously in focus, ensuring they are addressed even if time for other matters runs short. Communication plays a significant role, but it is also one of the most challenging aspects.

> *"Nothing we have discussed today works at any level without good communication."* (Interviewee, translated)

Communication must be able to succeed in all directions from all directions. It is essential for messages to flow between and within organizations, from top to bottom and bottom to top. This can be challenging, as the importance of the message may be downplayed, or it may change along the way. If communication between organizations relies on centralized methods, such as designated contact persons, it can cause delays and distortions in the communication chain. Therefore, the flow of information should be streamlined as much as possible to eliminate intermediaries and allow information to flow freely. This can mean, for example, establishing a dedicated discussion group where

stakeholders can exchange information with ease and personnel from different functions can communicate openly within certain rules and boundaries.

> *"The biggest problem with communication is assuming that it has occurred."* (Interviewee, translated)

Communication often plays an important role in conflict situations, and it is involved to some extent in all root cause analyses. According to one interviewee, concealing information or failing to disclose it, even when asked, is a frustrating factor in inter-organizational collaboration. This can stem from differing views on what information each party is allowed to discuss with the other. This further emphasizes the need to agree on common communication rules to ensure that each party receives the necessary information without unnecessary delays or distortions. Openness and transparency in communication are crucial for promoting a common safety culture.

4.2.3 Procurement activities and contractual terms

Considering safety culture during the partner selection process is one element influencing the overall quality and safety of inter-organizational collaboration. Often, product or service price and other financial factors are emphasized in bid comparisons, but according to the interviewees, emphasizing the vendor's capability and safety culture is just as important, if not more so in the long run. This underscores the perspective that considers the vendor's ability and willingness to produce a safe and high-quality product or service.

The importance of safety culture is particularly highlighted in safety-critical industries, especially in nuclear energy industry. Safety culture should be considered in the preliminary assessment of vendors or at the latest in the inquiry documents, where vendors are requested to provide their safety procedures and plans. This can include safety plans, but also more broadly documented practices and procedures to ensure that vendors are committed to a high safety culture. Inquiry documents, job descriptions, and delivery content should be as transparent as possible. These should also be reviewed together with the vendor to ensure they understand the nature of the collaboration and are ready to provide the required service or product in accordance with safety standards. This also requires a clear understanding of the client's own needs and requirements.

Potential pressure on partner organizations regarding safety issues begins with the procurement process. Expectations and standards are presented to potential partners, and meeting these standards requires consideration of safety in all activities. Each partner should

be regularly assessed to evaluate their capabilities and achievements in terms of safety and quality. These vendor assessments emphasize open communication and operational development methods that support the development and maintenance of a safety culture.

> *"Hard measures can motivate the development of safety issues much better than so-called nice-to-have requests, like 'it would be nice if you would do things safely'."* (Interviewee, translated)

As of today, safety considerations are mandatory in some industries. For example, in the nuclear energy industry, safety is a criterion that affects the overall score of a bid. Poor safety metrics lower the value of a bid, making safety aspects a competitive advantage. Safety improvements can also be encouraged with penalty clauses in delivery contracts if, for instance, occupational safety is neglected. This sets concrete financial incentives and responsibilities that force vendors to invest in safety.

> *"When such things are required, then real investment is necessary, and you have no other option."* (Interviewee, translated)

In traditional inter-organizational collaboration agreements, it is important to discuss common goals and standards already at the procurement stage. This ensures that all parties understand safety-related expectations. This helps ensure that collaboration proceeds smoothly, and safety issues are addressed at every stage. When developing collaboration agreements, it is important to plan joint efforts and collaboration carefully in advance. This helps avoid problems and blaming culture, which may be more common in inter-organizational collaboration than within a single organization. Setting up clear goals and milestones in the agreement can help reduce blame and improve collaboration, especially if the project is delayed or other issues arise. Regarding safety issues, it is useful to define what the collaboration will include and how it will be implemented during the contract development phase. This also concerns information transparency: clauses or commercial terms can be added to the contract to encourage openness in handling and reporting errors. This ensures that mistakes are brought to light and addressed appropriately instead of being discovered and corrected later.

> *"In our organizations, we are exploring collaborative project delivery models instead of traditional EPC/EPCM (engineering, procurement, construction / construction management). We are also examining how the alliance model could be utilized in massive industrial construction projects. Through business solutions, the alliance or other collaborative project management models are solutions to facilitate inter-organizational collaboration more effectively."* (Interviewee, translated)

Various collaborative project delivery models can significantly impact the safety culture in inter-organizational collaboration. Considering safety culture during the partner selection process ensures that all parties are committed to high safety standards. Collaborative project management models, such as the alliance model, can significantly improve safety culture. In these models, people from different organizations work together towards the same goal, creating a sense of unity and improving collaboration. In the service business, projects often rely on subcontractor networks, and practices supporting collaboration are not always sufficiently considered.

4.2.4 Leadership by example

The formation and development of a safety culture can be significantly influenced by setting a personal example. This requires that top management genuinely commits to safety, communicates its importance, and implements concrete actions that demonstrate safety as a top priority. When top managers lead by example through their actions, it sets a standard for the entire organization and encourages employees to adopt the same attitude.

> *"Top management must demonstrate through their concrete actions that safety is important in the organization. Mere 'safety first' slogans are not enough."* (Interviewee, translated)

To maintain or develop safety culture, it is crucial for management to also address situations where safety might be compromised. For example, under significant time pressure, leaders must take the extra pressure and responsibility off the operational level to ensure safe and quality operations. This is also one concrete leadership practice to influence other organizations. Interviews highlighted additional situations where it was necessary to refuse solutions that were morally and safety-wise questionable. In these cases, it is essential to explain why safety is a primary value and emphasize that the organization does not

compromise on its own standards. Commitment to safety can also mean not progressing into following phases until the safety of previous phases has been proven. This approach can also communicate to partner organizations what the priorities of one's organization are and demonstrate how safety-related concerns are truly primary, even in challenging situations.

> *"A partner in collaboration must be able to communicate to the other that safety is the primary priority, even if it feels like taking a business risk. This is exactly what safety culture is all about."* (Interviewee, translated)

These examples can communicate both externally and internally that safety is a primary value and a goal of collaboration. The organization's values and communication can broadly highlight their safety culture and require high standards from others as well. Organizations must adhere to their minimum requirements regardless of what customers expect. This sets the bar high and encourages other organizations to improve their practices.

4.2.5 Continuous improvement

By applying the principles of continuous improvement and learning to be a learning organization, companies can continuously adapt, develop, and enhance their safety over the long term. Neither the organization nor its management should become complacent with a sense of safety, even if things appear to be going well at the moment, because maintaining the current state requires effort.

> *"We should not be content with being good enough; we should always strive to improve. Just as top athletes do not stop training after winning a championship, they must continually improve their performance and stay actively engaged in practice to prevent their results from deteriorating."* (Interviewee, translated)

Safety culture can be proactively developed based on observations and suggestions reported within the organization. For example, continuous development and refinement of organizational guidelines based on practical experience and feedback help ensure they are clear, easy to follow, and effective. The iceberg phenomenon aptly illustrates how addressing small issues can prevent larger problems. When small mistakes are addressed and brought to light, they can be corrected before leading to more serious issues. It is important to encourage reporting and emphasize that no issue is too small to report. All errors and observations should be properly addressed to learn from them and avoid more serious cases

in the future. Ignoring reported issues can be extremely detrimental to safety culture, as it can lead to employees losing trust in the organization's systems, thereby weakening the safety culture. Continuous improvement often requires root cause analysis to understand why something happened and learn how to prevent similar occurrences in the future.

Benchmarking and learning from others are also significant methods for developing safety culture. Reflecting on the organization's current state and comparing it with other entities helps identify the strengths and weaknesses of its operations. Safety culture surveys and systematic measurement provide valuable information that can be used for benchmarking against other organizations. Adhering to established industry design standards and best practices, as well as leveraging good practices from other industries, can improve safety culture.

Collecting material and evidence from past events, their analyzed root causes and potential resolutions to form lessons learned material is a crucial part of a learning organization. These can be reviewed as educational material with the members of own organization and partners to learn from previous mistakes and improve practices. It is important to handle issues anonymously and as teaching stories rather than searching for culprits. This way, both the individual who made the mistake and the organization can learn from the event and avoid repeating similar errors.

4.2.6 Induction, orientation and training

Induction, orientation and training are widely used practices for influencing employees' attitudes and approaches to safety culture. They increase awareness and understanding of safety issues, which in turn strengthens the organization's safety culture and improves overall workplace safety. When employees understand why certain safety practices are followed, they are more likely to be motivated to adhere to them. This deeper understanding also allows employees to identify and highlight potential errors and improvement suggestions, leading to the continuous development of safety methods.

Safety culture training is one way to embed safety culture throughout the organization. For example, safety culture training organized for the entire organization helps instill safety matters from top management to all supervisory levels and other employee groups. Regular and continuous training and communication about organizational issues are especially important in larger organizations where employee turnover is high. Inducting new employees into the organization's desired practices ensures that practices remain consistent and that all employees understand their importance.

Furthermore, the organization may have a designated responsible team or person who can conduct safety training for new or external employees. The purpose of the training is to provide a basic understanding of what can and cannot be done and how to act in different situations. This ensures that new employees understand and follow the organization's practices from the start. External employees with their own cultures and procedures benefit from induction training, which helps them understand how to operate safely within the organization, such as in a nuclear power plant.

> *"What would be a meaningful and motivating way to bring up safety issues? Those heavy mandatory courses held once a year, like code of conduct reviews, may not be the way, but short and concise 'safety minutes' with a good agenda and real examples can be."* (Interviewee, translated)

To increase motivation and bring observations to light, meaningful and motivating methods can be used. Instead of traditional heavy mandatory courses held once a year, it may be more effective to organize tailored briefings for the target audience, dealing with real examples and concrete cases. These short sessions keep safety issues continuously in focus and help employees internalize them in their daily activities.

4.2.7 People involvement

Actively involving people in forming common goals, developing practices, and raising issues can significantly improve an organization's safety culture. When individuals are given the opportunity to influence these aspects, they feel like an important part of the organization and their perspectives are valued. This process leaves an experiential memory that their opinions matter and that their suggestions are heard and implemented concretely. This can increase their commitment and motivation to adhere to common safety practices or understand them more deeply.

The importance of involvement is emphasized already during the contract development phase of inter-organizational collaboration, where representatives of all parties should be included in defining what the collaboration will entail and how it will be implemented. This provides a better foundation for a common understanding and clear goals for the collaboration. Group discussions and soliciting members' opinions are also important ways to engage people. Encouraging group members to share their views and ideas promotes more active participation. Younger members can bring fresh perspectives, while more

experienced members can share their accumulated knowledge, creating a diverse and rich discussion culture. Safety rounds, inspections, and observation reports are concrete ways to participate in developing a safety culture. When all members of the organization, from management to employees, participate in these processes, safety becomes part of everyday operations.

> *"After the organization's management required to conduct safety walks, it became like a second nature to always check emergency exits, identify hazards, and report them when entering a worksite."* (Interviewee, translated)

Organizational learning often starts at the individual level. In smaller companies with a flat hierarchy, anyone can easily raise issues, and they are addressed quickly. This creates an open culture where everyone can participate in improving safety. In larger organizations, the process can be more complex, but the principle of involvement is the same: people's views are listened to, and they are encouraged to bring forward improvement suggestions. Involvement creates an open and transparent culture where everyone's perspectives are valued.

4.2.8 Rewarding

Influencing safety behavior through rewards is common in many industries but developing an effective reward system can be challenging if the metrics used can lead individuals toward questionable solutions. Safety performance can be measured, for example, by accident frequency and the incidence of severe accidents, but these metrics can be manipulated to appear better than they are. Interviews highlighted examples where a more serious accident was reported as if it had occurred outside the workplace to make the statistics look better. Additionally, in some organizations, safety issues can be seemingly taken to the extreme, but if a task involves even the slightest risk, they can outsource it to some external contractor. These practices do not improve overall safety but create a distorted view of the safety level or shift the risk elsewhere. Thus, safety metrics should consider both internal and external parties to be more transparent and reflect the real situation better.

> *"In one organization, safety issues were taken to the extreme, but if a task involved even the slightest risk, it was outsourced so that the company's own employees didn't have to take the smallest risk. At least their accident statistics stayed clean if something happened."* (Interviewee, translated)

It should also be noted that financial incentives tied to the number of accidents or other incidents can create improper social pressure, leading to dishonest reporting of events. Generally, incentives should focus on reporting errors or areas for improvement, promoting an open and honest safety culture. To motivate and commit top management to safety issues, short-term incentives have included EHSQ aspects. However, these aspects should generally be seen as essential for the company's long-term financial success, highlighting the strategic benefits they enable.

> *"Financial incentives should focus more on rewarding the reporting of errors or areas for improvement rather than the number of accidents because this encourages openly bringing up issues without fear of losing bonuses."* (Interviewee, translated)

Partner organizations, especially in project-based collaboration, can be motivated with milestone-based incentives aimed at achieving set goals more efficiently. This approach has been found effective, but challenges usually arise if milestones are not met within the agreed schedule. Additionally, penalties have typically been used for motivation, but positive incentives are a better approach; offering a carrot instead of a stick emphasizes positive actions. Motivating individual employees has been challenging, especially if they are not paid overtime or their working time banks are already full.

4.3 Factors contributing to development of safety culture

Based on the interviews, several factors contribute to the development of a safety culture, including financial incentives, regulatory requirements, and organizational commitment. These factors help create an environment where safety is prioritized and continuously improved.

4.3.1 Top management commitment

According to the interviewees, commitment from top management to organizational issues is the most crucial factor in building and developing safety culture. However, securing such a commitment can be challenging, as safety-related matters are often among the first to face budget cuts under financial and scheduling pressures. The foundation lies in the willingness, desire, and attitude of the management to work towards a safety culture, and the concrete development of it requires resources, time, and money. This investment originates from top management, which determines the budget and resources allocated to the organization's investments.

> *"Everything starts with the willingness, desire, and attitude to act on it. It's very easy to write and talk in buzzwords, but concretely developing a safety culture, for example, requires resources, time, and money."* (Interviewee, translated)

> *"The most toxic thing for safety culture is when issues are reported and even listed for corrective actions, but they do not progress due to poor processes, lack of resources, or interest."* (Interviewee, translated)

> *"Engaging top management in safety related issues is very difficult because it's often one of the first areas to face financial cuts."* (Interviewee, translated)

Business factors significantly impact top management's commitment to safety issues, but leaders need to understand that safety and financial success go hand in hand. A good safety culture generally reflects in the quality, costs, and schedules of deliverables and other activities. A well-developed safety culture can lead to successful investment projects and help stay within or even under the planned schedule and budget. In safety-critical organizations, operational safety and quality are prioritized factors, while business schedules and profitability are considered secondary business drivers.

However, it is essential to recognize that in low-margin businesses, it can be challenging to get the organization’s management to spend money on ensuring safety and quality, especially if the company’s survival is at stake. This financial starting point differentiates many organizations in other industries from those operating in safety-critical sectors, which typically generate enough financial benefit to invest differently back into

themselves. Financial pressures, tight time schedules, and unfavorably defined reward systems can also encourage the concealment of problems, which may cause significant issues later on.

> *"Especially in low-margin businesses, it's challenging to get any top executive to spend money on safety, which is perhaps seen as secondary when the company's survival is at stake."* (Interviewee, translated)

Commitment from management to developing and maintaining a safety culture is also critical for protecting the company's reputation and brand. This commitment directly affects how well a company can prevent safety, quality, and other deviations. When an organization's safety culture is in place, the number of deviations and accidents typically decreases, reducing associated costs and improving productivity. At the same time, the company's reputation and brand remain strong and protected. A good safety culture is a significant part of the organization's branding and can be communicated to potential customers through marketing and the organization's values. Additionally, a high level of safety culture can give a company a competitive edge, improving its chances of securing deals. This serves as a direct incentive for service providers, who can see the level of safety culture directly impacting their business opportunities.

> *"You can lose your reputation only once, and it can be incredibly difficult to recover."* (Interviewee, translated)

> *"Reputational damage and general business acceptability are significant factors also in other industries than just nuclear industry."* (Interviewee, translated)

Slipping in safety matters can lead to severe consequences, which can, at worst, end the business. This is particularly emphasized in safety-critical industries, where production losses and reputational damage are significant risks. In these sectors, the financial losses and reputational harm caused by deviations are often sufficient motivators to ensure adequate investment in safety from the start. Top management's commitment to promoting a high safety culture is one of the most significant factors in creating a strong foundation for long-term success, regardless of the industry.

4.3.2 Cultural differences

Cultural differences between nations or organizations are a significant factor in the development of a safety culture in inter-organizational interactions. According to interviewees, understanding the cultures of different organizations is a major challenge, but simply recognizing and acknowledging these differences can go a long way without requiring changes. Learning about the differences between cultures and practices can reduce misunderstandings and improve the effectiveness of collaboration. National culture influences individuals at a personal level, so it is important to understand the characteristics of different cultures. When differences are significant, one might have to think about things in ways that seem illogical from their own cultural perspective.

Cultural differences can be observed in practice, for example, when visitors at production facilities receive personal protective equipment, but local workers do not adhere to the same occupational safety standards. Some cultures may exhibit more indifference towards safety issues, significantly impacting the development of a shared safety culture. Additionally, different countries and cultures have varying work methods, gestures, values, and concepts, which can affect the organizational collaboration. Some cultures may prioritize profitability over safety or be more prone to seek out and punish those responsible for abnormal events, sometimes even publicly humiliating them.

> *"If you interviewed our former contractor or their subcontractors about safety culture, the interesting thing is that the paperwork is all there. Everything looks top-notch or even better than in Europe, but the practical implementation... we only need to take a week-long tour there, and the amount of trauma..."* (Interviewee, translated)

Within a single industry, there can be significant differences in how high safety issues are prioritized by companies. External vendors often have their own cultures and practices, and the question is how they can adapt to another organization's culture. National culture is less influential than the stage of collaboration between organizations. For example, startups and large, established organizations may approach issues differently, which reflects in their work culture and practices. Within a single organization, there are also different subcultures, leading to various approaches, such as a tendency to be more agile in operations or decision-making than the organization is generally perceived to be. These subcultures should be acknowledged, as they are generally a richness for the organization despite the challenges they may pose.

> *"You can recognize the readiness when you face organizations with different levels of capabilities and maturity. Some are very aware, and they don't need the safety aspect emphasized, but the lower the starting level in the other organization is, the more it requires going through things. This is a similar issue with both external and internal organizations."* (Interviewee, translated)

An organization comprises individuals whose traits and attitudes form the organization's culture. This makes culture an interesting aspect to manage, and changing it is not simple if the need for change arises. Individuals' attitudes and their histories significantly influence the development of safety culture in organizations. Patterns of behavior and experiences adopted in previous organizations become deeply ingrained in individuals, making it challenging to influence these attitudes. Attitudinal problems, where safety issues are not considered important, can arise from differences in interpreting what safety even means and how it should be approached. Attitudes toward organizational guidelines often divide opinions, but it is crucial that everyone understands the seriousness of safety issues. Unfortunately, there will always be a small number of people who do not take safety seriously and come up with reasons why they don't need to follow the guidelines, such as believing they are too experienced. At worst, the realization of the importance of safety happens only after something serious happens to themselves or a close colleague.

> *"Patterns of behavior learned elsewhere become so ingrained in individuals that if someone has a 'couldn't care less' attitude, it is quite difficult to learn out of it."* (Interviewee, translated)

Continuous communication about the importance of safety-related issues and their regular repetition are key when attempting to change attitudes and embed safety issues. When issues are repeated often enough and effectively embedded, they will start to stick with individuals over time. However, it must be remembered that forcing issues can lead to phenomenon, where the original purposes of the actions become blurred, and things are done only for formality without genuine deeper purpose and understanding. Therefore, it is necessary to find methods that suit each organization and its cultural environment.

4.3.3 Psychological safety

Psychological safety within an organization is an important factor influencing individuals' attitudes and the overall development of a safety culture as it enables employees feel safe

expressing their concerns, mistakes, and suggestions for improvement without fear of blame or punishment. The level of psychological safety within an organization directly affects whether genuine discussions about mistakes and learning occur. Top management's attitudes, actions, and the tone used in communication play a central role. Without psychological safety, employees may not dare to voice their mistakes, which prevents the organization from learning and developing. It is essential to create a culture where mistakes are openly discussed and continuously learned from.

> *"This is related to the psychological safety present in the organization, influenced by management's attitudes, actions, and the tone used in communication. Without psychological safety, genuine discussions on these topics do not occur."* (Interviewee, translated)

According to the interviewees, it is very common in some organizations that when something goes wrong, the first reaction is to find someone to blame, which creates a negative impact on the environment. Instead of searching for culprits, energy should be focused on learning from the mistake and how similar situations can be avoided in the future. Developing such culture starts by acknowledging that even if employees follow all guidelines, mistakes can still happen. If an error occurs despite following instructions and acting in good faith, the employee should not be blamed personally. Instead, the focus should be on learning from the incident and improving processes to prevent future occurrences.

> *"The way it usually goes is that when something goes wrong, the first thing people do is look for someone to blame instead of figuring out why it went wrong. Heads have to roll - that's always the spirit."* (Interviewee, translated)

When handling human errors, it is important to maintain a professional attitude, and blaming and attacking the person who made the mistake does not foster psychological safety. The key is to create an environment where mistakes are not feared, but rather, actions are taken, and possible errors are learned from. When a mistake occurs, the entire organization's processes and practices should be examined: why the mistake happened and what can be done to prevent it in the future. This helps turn a blaming culture into a learning culture, where continuous improvement is the main focus.

4.3.4 Risk awareness in decision making

Risk awareness and understanding their consequences are challenging factors in decision-making. Decisions often have to be made without complete certainty of their outcomes, but the associated risks must be identified as comprehensively as possible and minimized effectively. Risk assessments provide valuable information to decision-makers who have to weigh what level of risk is acceptable to advance matters. Often, risk assessments can lead to decisions to take safer actions if the adverse impact of risk realization is too high. It is crucial to communicate the severity of risks to the top decision-making body so that sufficient resources can be allocated for risk management.

The conflict between safety and financial objectives is common in decision-making situations. Increasing safety and work quality brings additional costs, and the question often is whether the customer or end-user is willing to pay more for safer operations. Interviewees provided several examples of decision-making situations where the speed of operations was maximized at the expense of quality or safety, often leading to problems in later stages. Project schedule delays typically cause additional costs and production losses, often leading to decisions to speed up planning or extend workdays, which usually compromises work quality.

> *"Decision-makers often find it challenging to see the long-term costs because they do not have time to conduct separate cost calculations between 'safe' and 'unsafe' solutions."* (Interviewee, translated)

Additionally, decisions made in a hurry can cause safety risks that decision-makers do not always recognize. Without a strong safety culture and understanding, assessing the consequences and costs of adverse events is challenging, and they may not be seen as sufficiently significant. Describing and explaining risks concretely often helps decision-makers understand them better, but sufficient time is often not allocated for this. Ensuring the safety of hydrogen production is an example of challenging decision-making, where technical solutions aim to reduce risk to an acceptable level. However, decision-making requires balancing costs and potential savings when discussing the acceptable level of risk relative to the additional costs of reducing it.

> "*Sometimes decision-makers do not understand safety risks until they are explained to them concretely."* (Interviewee, translated)

Safety issues should be central to business decisions and should be considered and incorporated into the decision-making process from the outset. Interviewees emphasized that quality standards must not be compromised, as this can lead to a snowball effect where quality deteriorates continuously. Over time, this can also erode safety-related matters if the overall quality level of the organization's operations declines significantly.

Decision-making typically becomes more conservative when safety-related issues are at its core. This is somewhat contradictory to the rapid growth objectives of growth companies, which creates its own challenges, for example, in financial arrangements. It is important for the organization to emphasize safety and ensure that all possible risks and their consequences are considered in decision-making. Interviewees also shared the common view that risk-based safety thinking is increasingly emphasized across all industries and players, and its importance in decision-making is continuously growing.

4.3.5 Regulatory framework

Regulation and legislation can significantly influence the consideration of safety issues, even to the point of compelling organizations to operate safely. Regulations can impose specific obligations that force organizations to invest in safety. While financial factors can sometimes be the focus of tough decision-making, when obligatory requirements come into force, organizations have no choice but to comply with the regulations. This ensures that safety issues are taken seriously enough.

> *"I don't believe that especially small and medium size enterprises would start spending money on safety out of goodwill because it would be 'nice-to-have'; it needs to be regulated instead."* (Interviewee, translated)

In Finland, business operations outside the nuclear energy sector are generally straightforward, and the regulations in these industries typically do not pose significant challenges to business organizations. Generally, current legislation defines safety requirements related to construction activities, which do not require significant preliminary safety preparations or paperwork. The safety of hydrogen production is a good example of regulation focusing on ensuring process safety, mainly requiring the presentation of appropriate technical protective solutions. In contrast, the nuclear energy sector's legislation and other official guidelines include multiple regulatory levels that define various requirements for both nuclear safety and the safety culture within organizations. However,

industry-experienced interviewees viewed the ongoing changes to Finland's nuclear energy regulations positively, seeing them as an opportunity to attract new players to the industry.

> *"It's a nice thought that every top business manager would think it would be great if all employees went home healthy, but it doesn't always work that way. It doesn't work just thinking with soft values; there must be some hard regulations behind it."* (Interviewee, translated)

Interviewees emphasized that motivating safety issues through regulation is perceived as bureaucratic and very slow, but often necessary. Other factors, such as financial losses and reputational damage, should be sufficient motivators, especially since in safety-critical sectors, the financial losses and general reputational damage from production losses are significant factors. These factors also extend to industries other than nuclear energy sector. However, regulation can ensure that a minimum level of consideration is taken so that organizations with weaker safety cultures do not pose too great a risk to their environment. Interviews also highlighted that the general tightening of the security climate due to recent global political situations has led to stricter requirements, potentially limiting collaboration opportunities with some customers based on national security requirements. This should be considered if an organization aims to conduct international business.

4.4 Reflection with the three levels of culture model

According to Schein's three-level culture model (1985; 2017), any culture can be studied through artifacts, espoused beliefs and values, and basic underlying assumptions. While artifacts are easily recognized, their meaning requires analyzing deeper levels, such as espoused values and beliefs, to identify underlying assumptions. When analyzing and reflecting on the contents of the interviews in relation to the theory adopted by the IAEA (2002b, p. 3-14; 2016c, p. 3-5), the following cultural elements were recognized. These elements can be seen representing the safety-critical industry culture as discussed for instance by Reiman & Oedewald (2008) and Viitanen (2015).

Artifacts

- Work accidents, production shutdowns, and other abnormal events
- Office, site, and production facilities
- Tools and personal protective equipment
- Performance indicators of the organization's operations
- Observation rounds and reports
- Risk analyses
- Permanent and temporary organizational structures
- Behavior and appearance of individuals
- Management system and certificates
- Content and quality of documents
- Organization's brand and reputation
- Training programs
- Meeting agenda
- Reward practices

Espoused values and beliefs

- Safety is the primary priority in business.
- A good safety culture provides a stable foundation for profitable business.
- Understanding and managing culture is complex but extremely important.
- Operations and communication must be as open and transparent as possible.
- It is possible to influence the values guiding the actions of organizations.
- More can be achieved through collaboration than alone.
- Attitudes towards safety issues vary between countries, industries and organizations.
- Leading by example in safety aspects may require taking considerable business risks.
- Maintaining and improving performance requires continuous effort.
- Poorly done things and mistakes cannot always be corrected afterwards.
- Mistakes can and should be used for learning opportunities.
- Regulation is necessary to protect the environment from the hazards caused by organizations taking business risks.

Underlying assumptions

- In many industries, top management may focus too much on financial performance at the expense of other business matters.
- Business continuity is more essential for many than its general acceptability.
- Abnormal events are often caused by negligence or a lack of understanding.
- It is always possible to improve performance and quality of operations.
- Individuals are willing to question things for the common good.
- Organizations and individuals fundamentally pursue only their own financial interests.

These identified elements of different levels of safety culture are interpretations of the issues discussed in the interviews which can be considered to be related to the research context. Artifacts are observable elements that emerged in the interviews, while espoused values and beliefs are those things the interviewees themselves mentioned or described as being important. Underlying assumptions are elements or concerns that were not directly stated or described in the interviews but can be interpreted to be present from the interviewees' responses.

These lists are not comprehensive, and they could be expanded, and categorized according to different characteristics of safety culture. It is also important to note that the purpose of these listings is not to evaluate personal values of the interviewees nor the cultures or business objectives of the organizations they have presented before or at the time of the interviews. These are author's personal interpretations of the artifacts, elements, espoused values and beliefs, and underlying assumptions that may generally exist in safety-critical industries and the organizations operating within them.

5 Conclusions and discussion

This chapter presents a comprehensive conclusion of the main findings from the study, comparing the them to the reviewed literature to answer the research questions. To address the broader research theme of safety culture development, actionable suggestions for further developing safety culture are introduced, considering both individual organizations and business networks by integrating cultural and organizational development insights from this study. This is followed by the presentation of ideas for future research, pointing out potential areas for further study in the field of safety culture in the business operation context. Finally, the chapter discusses the study's limitations and reflections, highlighting the challenges and general observations encountered during this study.

5.1 Main findings and answers to research questions

5.1.1 Appearance and recognition of safety culture

To answer the first research question on *how safety culture appears in business networks and how it is assessed*, the findings reveal few key points on how safety culture needs to be systemically integrated into the organization and its business strategy. According to the findings, it can be concluded that a strong safety culture integrated into the operations at all levels of an organization forms a solid foundation for sustainable and profitable business.

The strategic importance of organization culture and systematic approach for safety culture were clearly recognized as discussed by the IAEA (2016a; 2020) and Gotcheva & Oedewald (2015b), among many others. The appearance of safety culture may vary by industry and organization, but they share the same fundamental principles and elements for the creation and development of a safety culture. These findings align with those of Williams (2017), Ahola et al. (2014), and Ruuska et al. (2011), who suggest that certain cultural aspects may correlate with the financial performance of the organization, although verifying this is challenging.

Comprehensive assessment of safety culture requires a holistic approach, combining metrics, surveys, and audits. In here, reactive indicators provide essential information about past events, while proactive indicators are essential for the prevention of future hazards. These views align with commonly used assessment frameworks as discussed for instance by the IAEA (2002b; 2009) and Reiman & Oedewald (2008). However, only through surveys and audits, it is possible to obtain a comprehensive picture of the organization's safety culture level and ensure the truthfulness of documentation related to the organization's operation, as

is discussed also by Reiman & Oedewald (2008). This is important to acknowledge when considering and planning collaboration with external organizations, especially those from national cultures often associated with unsatisfactory safety culture.

5.1.2 Practices to influence elements of safety culture

To answer the second research question on *how elements of safety culture can be influenced in practice*, the findings reveal several key practices. These include fostering a unified understanding of shared goals and ambitions, establishing commonly agreed ways to collaborate and communicate, incorporating safety culture aspects and safety performance into business partnerships, leading by example, implementing continuous improvement into organizational processes, developing proper training frameworks, involving people in decision-making and development processes, and creating sustainable reward systems.

Creating shared goals and a unified vision forms a strong foundation for collaboration across organizational boundaries. Additionally, mutually agreed practices and communication methods help ensure that all parties understand each other, enabling information to flow without delays and distortions. This openness and transparency in operations and communication increase trust both within individual organizations and between them. These are commonly recognized practices to align ambitions between organizations (Cummings et al., 2019, p. 510), while ensuring operational effectiveness within their collaboration (IAEA, 2002a; INSAG, 2002; Milch & Laumann, 2016). However, as noted by Oedewald & Gotcheva (2015b, p. 112), leadership in networks is often distributed and incoherent with many managers and leaders holding diverse perspective, which can make it challenging to establish a clear vision and the practical steps needed for safety culture development.

When selecting a business partner, it is essential to emphasize the organization's ability and willingness to understand and commit to high safety and quality standards as were also emphasized by Cotcheva (2016). The business partner selection and procurement process should evaluate their capabilities and achievements from safety and quality perspectives where the price of the product or service should not be the sole determining factor in decision-making as long-term impacts of the collaboration should also be considered. For instance, these may include reputational and public acceptability factors of the business which can be difficult to recover (Reiman & Oedewald, 2008, pp. 5-6). In addition, inter-organizational operations can be enhanced through networked organization structure

(Ruuska et al., 2011), or with collaborative project execution models, where collaboration is closer and more transparent than in traditional approaches.

The formation and development of a safety culture can be significantly influenced by the example set by management and the concrete actions they take to demonstrate the high priority of safety within the organization (IAEA, 2020; Reiman & Oedewald, 2008, p. 47). Only by acting in the desired manner can they expect and encourage individuals and partners to adopt the same attitude. If communication and actions by the management are disconnected, it sends a confusing message to others about the desired practices and priorities. (Schein, 2017, pp. 188-189).

To keep pace with general development and change, the organization must systematically improve its operations and learn from abnormal events and other experiences. This emphasizes that organizations should become '*learning organizations*' (INSAG, 2002, p. 10; Cummings et al., 2019, p. 490-491), where priority should be placed on proactive measures for creating an encouraging environment for learning, even though reactive learning is also necessary. Key methods for organizational learning include collecting ideas and feedback, as well as benchmarking and lessons learned approaches, which involve learning from other organizations and identifying the root causes of abnormal events (INSAG, 2002, pp. 14-15; IAEA, 2020).

When considering individual members of the organizations, a good way to influence safety behavior and attitudes seems to be through orientation and training (Reiman & Oedewald, 2008, p. 37), as well as involving them in development and decision-making processes. When individuals understand the significance of existing practices and are aware of the underlying factors, they are more committed to following these practices (Roberts & Geller, 2018, p. 696). Participation also enhances openness and transparency, which can build trust towards the practices and systems of the organization.

Rewarding and punishing were also identified as possible ways to influence the behavior of organizations and individuals, but they were found challenging to implement to achieve the desired effect. However, it is important to note that rewards create a more positive foundation for collaboration compared to punishments, such as fines or penalties. (Schein, 2017, pp. 175-176). Additionally, rewarding does not predispose to a blaming atmosphere in the same way, although incentive programs often have negative effect on reporting (Viitanen, 2015, p. 39). Therefore, it is important to use metrics in reward systems that promote an open and honest safety culture and reflect the true operational situation.

5.1.3 Factors contributing to development of safety culture

To answer the third research question on *which factors contribute to the development of safety culture*, the findings suggest that top management commitment is the most critical factor. Other recognized factors include cultural differences between organizations and nations, the presence of psychological safety, the risk-awareness of decision-makers, and the regulatory framework in force that sets requirements or expectations for business operators.

The commitment of top management is the primary prerequisite for the development of a safety culture which was emphasized even more than organizational capabilities for continuous improvement as were discussed, for instance, by the IAEA (2002a, pp. 9-10), the INSAG (2002), and Oedewald & Gotcheva (2015b, p. 113). This is further supported by concerns that the financial condition of the organization and other business foundations significantly influence the level of top management commitment and the available options. Developing a sustainable safety culture requires the ability and understanding how to balance organizational performance, profitability, and other business objectives. In safety-critical environments, ensuring safety is a prerequisite for conducting business in these sectors, as the general acceptability of the business and the organization's reputation can affect business continuity. Therefore, it is not just an operational issue but also a strategic factor that directly impacts the organization's ability to succeed.

It was also recognized that business objectives and the financial pressures they create can influence decision-making, as noted by IAEA (2002a, p. 21) and Milch & Lauman (2016), where direct or indirect risks related to the decision may not always be adequately considered due to their sociotechnical complexity (Oedewald & Gotcheva, 2015b, p. 115). Additional resources needed for identifying or mitigating risk factors may be seen as unnecessary, too costly, or time-consuming, potentially impacting profitability or competitive positioning. Sometimes, the risks associated with decisions are not communicated clearly enough for decision-makers to understand them adequately, leading to decisions made with insufficient understanding within budget and timeline constraints dictated by the business objectives. These findings align with the decision-making biases discussed by Reiman & Oedewald (2008, pp. 74-80). These issues seem to be quite normal in business risk-taking, especially in startup or non-safety-critical business environments. However, in safety-critical environments, it is essential for the organization to emphasize risk awareness, which typically makes decision-making more conservative (Reiman & Oedewald, 2008, pp. 207-208).

Cultural factors at national, industry, and organizational levels also affect the development of a safety culture (IAEA, 2002a, pp. 21-22), as they may influence how willing an organization is to learn and how they value safety, quality, or profitability (Reiman et al., 2008, p. 21). The cultures in which individuals have worked and grown up also significantly influence their learned behaviors, attitudes, and assumptions. These characteristics are not easy to change once they are deeply ingrained. (Schein, 2017). Therefore, it is essential to understand and consider cultural differences to build a strong and unified safety culture in inter-organizational collaboration (Milch & Laumann, 2016). However, these differences can also make it more challenging to achieve certain safety culture characteristics in some national cultures. In organization networks, diverse cultures persist despite safety culture initiatives, but understanding national cultural dimensions (Hofstede, 2010, pp. 29-31; Williams, 2017, pp. 169-171), and management styles can help prevent misunderstandings and facilitate effective working practices (Oedewald & Gotcheva, 2015b, pp. 113-114).

Psychological safety within organizations is an important factor which should not be underestimated. It enables proper communication and information sharing inside the organization, and it is therefore important ingredient for the development of safety culture. Organizations should strive to create an environment where individuals and other partner organizations can present their ideas, mistakes, or concerns without fear of blame, embarrassment, or other negative consequences. This requires a generally encouraging attitude and mindful communication, where the gestures and tone used by management are impactful (Schein, 2017, p. 172). Furthermore, similar phenomena can be recognized to exist also on the inter-organizational level, as by admitting safety or quality deviations may cause a threat of being replaced or excluded from future co-operation opportunities (Oedewald & Gotcheva, 2015b, p. 112).

When organizations engage in commercial activities, free market-driven guidance should be a sufficient motivator to guide their operations into desired direction. In such cases, customers and partners using their products and services should influence the market offerings through their choices. However, investing in safety should not be seen as just a '*nice-to-have*' aspect but as an essential part of conducting business, especially in safety-critical industries. This makes regulation and oversight necessary to ensure that organizations invest sufficiently in safety to prevent unnecessary danger or risk to their stakeholders, environment, or other assets (Grote, 2018). Thus, regulation could directly target the operational activities of business operators by setting direct requirements or

indirectly through criteria related to financing. However, this bureaucratic and top-down approach may not necessarily trigger the proper internal response needed to create sustainable safety culture (Oedewald & Gotcheva, 2015b, p. 116), as it may lead to mere rule-compliance rather than genuine self-awareness. Over-regulation could even lead to attempts to find loopholes to navigate through the bureaucracy, which is not the intended purpose if regulatory measures are correctly placed. Therefore, organizations should be encouraged and guided to independently assess and implement necessary activities to ensure the long-term sustainability of their operations, including safety aspects.

5.2 Implementation actions for safety culture development

In addressing the broader research objective concerning *how safety culture can be developed within safety-critical business networks*, the main findings of this study, alongside the organizational development intervention literature discussed in section 2.3, offer a solid foundation for practical implementation actions and considerations for both individual organizations and business networks.

First of all, organizational cultures are very difficult to change, and there are no silver bullets to get around this challenge. This is even more challenging in network organizations due to the practical difficulties in crossing organizational boundaries (Oedewald & Gotcheva, 2015b, p. 120). Developing safety culture aspects requires strategic planning, long-term commitment, sufficient resource allocation, and significant time for desired changes to be realized, allowing individuals and organizations to adapt. The estimated time to see benefits from these changes can vary up to several years, depending on the complexity of the change and the size of the organization (IAEA, 2002a, p. 13).

Obviously, some quick fixes can be done in a more accelerated time scale, but to really make them stick and remain in the organization requires time and additional efforts for the culture behind behavioral aspects to change too. There are many practical examples of organizational improvement projects aimed at creating new practices and processes, or changing existing ones, but over time they tend to vanish away as old established behavior persists. Such outcomes may indicate lack of planning, resources or commitment, or resistance to change as an organization is not able to unlearn its embedded routines (Schein, 2017, pp. 325-326). The best results can be obtained when multiple approaches to develop organizational excellence are combined, including changing visible cultural artifacts, substituting new behaviors for old ones, and hiring individuals whose values align with the

desired culture (Willians, 2017, p. 65). Therefore, longer-lasting effects can be achieved with properly planned and executed planned change.

One potential challenge for many publicly listed and traded companies may be their focus on quarterly results and short-term performance-based rewards. These financial and business pressures can potentially distract management from their commitment to safety culture (IAEA, 2002a, p. 21; Milch & Laumann, 2016). Additionally, financial constraints can limit an organization's ability to invest in their organizational development. These can create a culture where immediate financial gains are prioritized over long-term initiatives. To address this, organizations need to align their reward systems with metrics that value long-term success rather than short-term financial results. They can also prioritize low-cost but high-impact development measures until their financial status stabilizes.

5.2.1 Individual organizations

Entering and contracting

Planned change begins with entering and contracting phase, where main target is to gather initial data to understand organizational problems or opportunities, and to help management of the organization to decide whether to proceed with a change program and commit resources to it (Cummings et al., 2019, p. 28, pp. 70-76). Before initiating a development program to achieve the desired change in safety culture, it is crucial to secure management commitment at all levels and ensure that sufficient resources are available. Integrating safety culture aspects into the business planning process can also emphasize the business importance of safety culture (IAEA, 2002a, p. 13). For example, following questions, but not limiting into, could be used for guidance to understand the initial situation from organizational perspective:

- What do safety and quality mean in the organization's business environment?
- How can the business strategy and strong safety culture support each other?
- Does the organization have a clear vision about its business in the future?
- Is the organization capable and willing to invest in improving its operations to maintain long-term competitiveness?
- What kind of brand image does the organization want to create for its stakeholders?
- Does the organization have experience of previous or existing operational challenges, and what has it learned from them?
- Have there been any changes in the organization's internal or external operating environment, or are such changes anticipated?

By answering the above questions, organizations should get some level of understanding how important safety, quality, and continuous improvement are for them in their own business environment. As pointed out by the interviewees, safety culture and business strategy have a major influence on each other and integrating them is essential for organizations operating in safety-critical industries. Among other discussed benefits, a strong safety culture enhances an organization's reputation, making it more attractive to customers, partners and employees. Proactivity and organizational learning are key elements to create a culture which can anticipate future challenges and issues, and to continuously improve its operational processes.

Diagnosing

After the initiation phase, the organization diagnoses the causes and consequences of its operational issues by gathering and analyzing information from appropriate sources. Culture change interventions generally begin by diagnosing the existing culture to assess its compatibility with current or proposed business strategies (Cummings et al., 2019, p. 28, pp. 83-92). According to the IAEA (2002b, p. 33) diagnosing organizational issues often reveals a gap between how things are discussed and their actual state. For example, teamwork may be valued but rewards are based on individual competition. To understand deeper organizational culture, it is needed to link espoused values with visible artifacts and identify inconsistencies and conflicts in them which helps to uncover the deeper assumptions. Recognizing patterns among basic assumptions driving the system helps explain the presence of various artifacts. Additionally, in case organization may consist of cultural sub-groups, they may have their own distinct assumptions which need to be considered. To understand the culture of an organization, diagnosing phase should follow these steps (IAEA, 2002b, pp. 32-34):

1. Identify artifacts: Observable elements and documented processes that provide visible evidence of the organization's culture.
2. Identify espoused values and beliefs: The public statements that the organization claims to hold.
3. Compare espoused values with artifacts: Analyze discrepancies between what is said stated to be and what can be observed.
4. Deduce basic assumptions: Uncover the underlying assumptions that influence organizational behavior and decisions.

By comprehensively analyzing findings, or by using a simplified method shown in Appendix B, organizations can assess the current level of their safety culture which the IAEA (2002b, pp. 17-19) has classified into three development identifiable stages as introduced in section 2.2.4 of this thesis. These findings and analyses can then be used to discuss and plan development areas, desired outcomes, and activities to achieve them.

Planning and implementing

In following the planning and implementation phase, the organization designs development interventions to address identified issues and to fit with its cultural and business context by considering its existing cultural state, readiness for change, and capabilities to execute it (Cummings et al., 2019, pp. 28-29, pp. 146-147). During the planning, it is important to consider the time required for cultural development and set a series of milestones to track progress. These milestones do not only help monitor actual progress but also motivate people during the long-term implementation, as intermediate targets can be shown to be achieved (IAEA, 2002a, p. 13). Additionally, it may sometimes be necessary to acquire new change management skills or appoint a dedicated change management leader to successfully implement the planned interventions.

The scope of the development intervention can range from a complete transformation of the entire organization to targeted changes within specific sub-units and subcultures. This influences the selection of the feasible approach for the intervention, which also depends on the organizational maturity stages on its life cycle: whether it is in a growing, midlife, or mature stage, as each one has its distinct change mechanisms that must be considered to effectively implement and sustain the desired changes (Schein, 2017). Regardless of the planned change scope, it needs to be clearly defined what needs to be changed and what the desired outcome is. According to analysis of the theory and the interviews, at least following elements should be considered in intervention planning to enhance safety culture of the organization:

- Sustainable reward system which considers both short and long-term aspects of the organization to motivate and encourage top management and individuals. Rewarding should be based on proactive measures and safety-focused development metrics to create a positive reinforcement loop. Rewarding based on only number of incidents and events should not be in a major role due to their risk to not support open and honest reporting.

- Operational processes, routines and communication should be open, transparent, and traceable which enables the organization to share knowledge effectively, introduce newcomers to use them, and to form a basis for organizational learning. Additionally, the type, language, and level of detail in the information provided for decision-making purposes need to be carefully considered to ensure that decision-makers are able to understand the consequences of their decisions. In mature organizations these are often in a documented format as part of their management system, and under periodical evaluation and development.
- Implementing measures to collect and process feedback is essential to gather insights and to continuously develop operational activities. Organizations should additionally consider implementing benchmarking and lessons learned practices to learn from others and identify root causes of events for corrective actions.

While some improvements can be achieved more quickly, embedding these changes into the organizational culture requires long-term commitment and sustained effort. Thus, organizations should set realistic targets and timelines, understanding that achieving concreate benefits from organizational development interventions can take multiple years, particularly for more complex changes.

Evaluating and institutionalizing

The final phase of planned change, evaluating and institutionalizing aims to ensure that safety culture improvements are sustained over time (Cummings et al., 2019, p. 29, pp. 184-201). By integrating key practices such as sophisticated operational activities, feedback and improvement processes, and sustainable reward systems, organizations can embed long-term safety aspects into their core operations. Evaluating the results and effectiveness of interventions allows organizations to make informed decisions about necessary adjustments. This ongoing assessment ensures that the improvements remain relevant and effective over time. The objective is to emphasize organizational learning through continuous feedback loops that facilitate knowledge sharing among its members, and the integration of best practices into organizational routines. This highlights that although organizational learning can be modeled as a continuous process, the journey is more like a spiral from the organization's perspective. Throughout its life cycle, the organization develops new aspects of its operations, storing these in its artifacts and ultimately embedding them in its culture. This approach has been identified by the IAEA in their guidance for safety culture development, as illustrated in Figure 5.

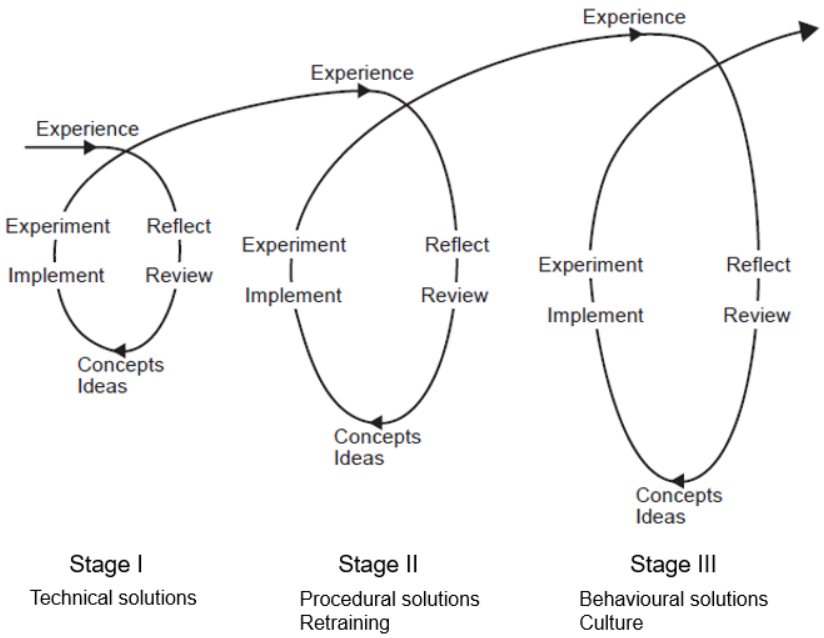


Figure 5. Simple model of safety culture development (IAEA, 1998, p. 9, modified).

Enhancing understanding and awareness of safety-related issues is essential for creating and maintaining a strong safety culture. This can be achieved through transparent communication and decision-making processes, which enhance an environment of trust and collaboration. Additionally, creating and sustaining a psychologically safe environment, where employees feel comfortable sharing ideas, mistakes, and concerns without fear of blame or retribution, supports this organizational learning process. Top management plays a key role in demonstrating these behaviors and committing to safety culture, ensuring the long-term success of safety culture development interventions.

5.2.2 Business networks

From business network perspective, a transorganizational change needs to be considered which involves interventions that extend beyond a single organization which can become considerable for example in case of networking organizations, alliances, or similar collaboration business operations (Cummings et al., 2019, p. 510). As discussed in section 2.3.4, these collaboration methods are used to manage tasks that single entities cannot handle alone, or to leverage each other's strengths for competitive advantage. They mainly differ in how formal the arrangement is, and how deeply their goals and interests are aligned together as networked organizations typically have self-interest in addition to the shared common interests (Cummings et al., 2019, p. 522). For example, following questions, but not limiting into, could be used for guidance to understand the initial situation from business network perspective:

- What business needs or opportunities requiring deeper multi-organizational collaboration exist in the near future? What do they require from the organization or potential partner organizations?
- With what types of entities does the organization want to collaborate in the future, and under what conditions?
- What kind of example and brand image does the organization want to set for other entities and stakeholders?
- How can collaboration agreements be structured to require higher safety and quality standards from other organizations?
- How can the organization encourage and motivate its partners?
- What collaborative business methods should be studied and adopted?

One significant challenge in transorganizational change is that an external organization's culture cannot and should not be directly changed; they need to change it themselves. Managing change in multi-organization business environments begins with partner screening and selection where organizational relationships and roles need to be understood among participating entities (Cummings et al., 2019, pp. 523-524). Deciphering differences between two companies in the same national culture is easier than deciphering both national and company differences in cross-national partnerships. Determining whether perceived differences are attributable to national or organizational cultures is essential since changing national characteristics is highly unlikely. (Schein, 2010, p. 379). This cultural diversity needs to be considered when assisting external organizations in developing their safety culture, and practices should be selected flexibly to achieve specific safety culture goals (IAEA, 2002a, pp. 21-22).

Possibilities to influence and motivate external organizations to change their practices, which could have a long-term impact on their culture, are limited and often focus on financial motivation through business incentives. According to analysis of the interviews, at least following elements should be considered in inter-organizational collaboration:

- Understanding the cultural fit between organizations by auditing partner candidates and comparing it to the safety culture of one's own organization. This enables the identification of possible conflicting areas and the evaluation of how they could be managed in future collaboration.

- Including safety culture aspects into inquiry and bid comparison processes and gradually increasing the weighting of it so that it would create the desired long-term influencing impact.
- Including safety related aspects in contractual agreements to ensure that partners understand the importance of safety and are committed to creating, maintaining, and enhancing high-level standards.
- Offering rewards and bonuses for achieving specific milestones or maintaining high safety standards can incentivize external organizations to prioritize safety.
- Including proactive improvement and people involvement aspects into bonus metrics by challenging partners to provide feedback and to continuously improve collaboration and processes. This could improve transparency of communication where safety related issues are brought to light instead of concealing them or denying their existence to avoid loss of bonuses.

In addition to the discussed business aspects, regulation and legislation can be used to force organizations to include or change certain aspects of their operations. For example, regulation and supervision in the nuclear energy industry have required business operators to consider organizational factors such as culture and leadership that affect their operations (IAEA, 2016b; STUK, 2019). According to views expressed in the interviews, regulation is one of the most significant means of influencing organization's perspectives on safety culture and its development, especially if the organization cannot achieve this independently. Additionally, financial incentives incorporated into legislation could effectively guide safety culture development.

5.3 Suggestions for future research

As the findings of this research indicate, financial incentives and regulation are major external factors which can motivate the organizations, and fundamentally its owners and top management, to consider safety aspects of their business operations. These financial incentive systems could be seen in three levels: a single organization, business network, and societal levels, where politics and regulation could be used as first incentive to guide organizations to develop their own business solutions to support safety culture development:

- From a single organization perspective: How would an internal reward systems look that encourages sustainable operations and motivates open and honest behavior within a single organization?

- From a business network perspective: How would a collaborational reward systems look that encourages sustainable operations and motivates open and honest behavior in external partner organizations?
- From society perspective: How could an organization's safety culture be considered in investment financing? How could safety culture related aspects be highlighted as part of sustainability reporting?

Investigating the impact of reward systems on top management and individuals can provide insights into fostering a safety-first mentality. Research could focus on identifying effective incentives such as performance-based bonuses, long-term incentives tied to safety outcomes, and non-financial rewards that recognize safety contributions. Understanding the behavioral impact of these incentives can help create a culture where safety is integrated into daily operations and decision-making processes.

External organizations' top management and individuals also play a crucial role in developing a cohesive safety culture across business networks. Research could examine how to incentivize external partners, such as technology vendors and service providers, to adopt similar safety standards. Financial incentives and performance-based contracts could be areas of focus. Investigating how these incentives align with different organizations and business models could provide valuable insights into fostering long-term safety commitments in inter-organizational business collaboration.

Furthermore, it could be investigated how safety culture aspects could be integrated into investment financing or funding criteria. By considering safety culture as a factor in investment funding, society can promote sustainable business practices and reduce risks associated with poor safety culture. Integrating safety culture into ESG (environmental, social, and governance) criteria could further emphasize its importance in sustainable business practices. Notably, the European Union’s CSRD (corporate sustainability reporting directive) already mandates the recognition of corporate culture and the inclusion of governance aspects in corporate sustainability reporting. Research could thus explore how to incorporate sufficient and reasonable safety measures into ESG assessments or corporate sustainability reporting, ensuring that safety culture is recognized as a key component of organizational sustainability. Such integration would drive business organizations to adopt sustainable safety practices and meet societal expectations.

5.4 Limitations and reflections of this study

One of the primary limitations of this study is the relatively broad and abstract research objective, which posed a challenge in maintaining focus. By narrowing and more accurately defining the research scope, a deeper exploration of certain topics could have been enabled. The desire to include actionable practices broadened the scope, which resulted in a more general level discussion of some aspects rather than an in-depth scientific analysis. This can be clearly recognized in the descriptive nature of the research questions rather than explanatory ones. The personal interest of the researcher in exploring how safety culture could be developed in inter-organizational collaboration may have influenced the choice of approaches and perspectives, despite the best efforts to maintain objectivity of this study.

This study utilized multiple compiled works and publications primarily focused on the nuclear energy sector, which also funds much of the research in this context. Due to the researcher's professional background on that industry, the specific characteristics, research entities, and literature of this sector were familiar, making them readily accessible, possibly without careful examination or looking at wider academic criticisms on them. Other examined literature on organizational and safety culture included many scientific publications addressing the human aspects of organizations. However, many of these articles focused on work processes, conditions, and attitudes of lower-level members of the organization, which were not particularly relevant to this thesis, as the interest shifted more towards the attitudes and motivational factors of top management. This is understandable, as occupational safety is the main safety concern for many industries and easy to recognize and understand, whereas safety culture related motivational factors behind business operations and upper-level decision-making may be too sensitive to be studied and revealed.

Organizational culture was primarily addressed using Schein's (2010; 2017) three levels of organization culture, a framework also applied by the IAEA in its safety culture publications. The works of Cummings et al. (2019) and Williams (2017) on organizational development and leadership were also utilized. While many other approaches and theories exist to understand the complexity of organizations, these relatively simple theories and models seemed suitable for this study due to their educational nature, and as they were already interconnected and mutually cited in the reviewed literature.

Analysis of the interview data revealed practical challenges and methods widely recognized in the literature. The underlying factors appear to be well-known to those working in safety-critical industries, which was not surprising. Including individuals without

experience in safety-critical organizations, or representing other national cultures than Finnish, could have provided more insights and perspectives. However, the selected target group offered valuable insights into the challenges in safety-critical industries, which could also be seen as expectations of current business operators for potential newcomers. The diversity of professions and experiences among the interviewees enriched the interviews, deepening and elaborating the guiding questions from multiple perspectives. Many interviewees mentioned that the anonymity of the interview allowed them to express their personal opinions and thoughts more freely, rather than the publicly expressed values and beliefs of their current organizations. Therefore, being disconnected from their current employers may have led to more critical opinions and perspectives on safety and business concerns than would otherwise have been possible.

This study identified practices and factors influencing the development of safety culture in safety-critical industries. According to the interviews, the biggest challenge is gaining top management's commitment and understanding of safety culture's impact on business operations. Concerns about the influence of commercial interests on top management's motivation were particularly evident. While this relation was recognized in the reviewed literature, this study suggests its significance may be greater than previously discussed. To conclude this study, one already introduced quote from the interviews punctually summarizes the role and purpose of safety culture in business network context:

> *"A partner in collaboration must be able to communicate to the other that safety is the primary priority, even if it feels like taking a business risk. This is exactly what safety culture is all about."* (Interviewee, translated)

Furthermore, an important consideration is that, as the IAEA (2016c, p. 27) proposes, the results of safety culture related studies should serve as inputs for organizational development processes. Therefore, the contents and outcomes of this study should primarily be used to build awareness and facilitate new ways of questioning how and why members of an organization across all levels act and think as they do.

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Appendix A: Common indications of safety culture

Table A-1. Examples of positive, or healthy, indications of safety culture (IAEA, 2002b; Oedewald, 2014; Reiman & Oedewald, 2008; Reiman & Rollenhagen, 2018).

Company recognizes societal, organizational and human elements as important for their safety and allocate resources and utilize expertise needed to deal with them.
Long term safety impacts of re-organizations, commercial contracts etc. major decisions are analyzed.
Workers actively aim for understanding their impact on safety; they show capabilities for analyzing safety effects of various decisions, practices and challenges.
Company uses multiple safety indicators, not only lagging indicators.
If workers fail in their job performance the work conditions and other prerequisites for good work are analyzed and improved.
Organization aims at identifying its cultural assumptions e.g. in event investigations or periodical safety assessments.
Safety policy, which includes the organization's vision, goals, and official criteria, as well as general principles against which operations are evaluated.
Participatory and democratic leadership style and visible commitment to safety by management.
Clear definition of responsibilities and obligations, including the clarity of job tasks and their safety significance.
Challenging unsafe behavior and a sense of responsibility for safety by everyone.
Competent risk assessment and awareness of risks related to operations.
Ability to adapt quickly to changing requirements and act flexibly and effectively in unexpected exceptional situations.
Positive employee motivation and job satisfaction.
Fairness, openness and trust among employees and management.
High-quality and up-to-date documentation and procedures.
Sufficient investigation, reporting, and learning from incidents and accidents.
Open communication and adequate collaboration across different levels and areas within the organization.
Continuous improvement of operations and safety.
Top management of the organization is committed to safety.
Questioning attitude among members of organization.
Whistleblowing culture exists.
Employees have a good understanding not only of their own work processes, but also how these processes interact with other processes.
Mistakes are seen as possibilities to learn.
The world is recognized as being complex and its understanding requires more than simple linear causal logic. Systems approach is being used to model it.

Table A-2. Examples of negative, or unhealthy, indications of safety culture (IAEA, 2002b; Oedewald, 2014; Reiman & Rollenhagen, 2018).

Safety is equaled to technical reliability or component quality.
Events are analyzed mainly from technical point of view and only quickly applicable corrective actions are suggested.
Organization applies rigor to deal with technical faults but skips "bureaucracy" as irrelevant: documenting work, updating instructions etc.
Workers whose work does not directly involve tasks with reactor or safety systems do not participate in nuclear safety trainings because "their work does not have safety significance".
Company uses occupational safety measures e.g. LTI as an indicator of nuclear safety level.
Management says they don't engage in safety culture work until safety culture can be measured objectively and in a quantitative manner.
Management assumes that activities are carried out exactly as they are prescribed in procedures.
Safety is given low priority in decision-making concerning, e.g., investments or production issues. Short-term profits dominate the management attention.
Safety is considered to be the safety organization's responsibility or something that is done only for the regulator. Safety function does not have the authority to influence operative or strategic decisions.
Human influence on safety is seen mainly as a source of errors, or it is not considered at all.
The organization glorifies its past and sees itself as already having developed all the risk control measures it needs. Safety is seen as a static state that has already been accomplished.
Many warning signals, alarms, or deviations are ignored since they recur so often; they have become normalized and considered part of everyday life.
Individuals and departments optimize their own performance without considering how that affects the overall performance of the organization.
The safety management system exists only on paper and does not guide actual practices. Procedures and instructions are not followed in real life.
The organization is in a fire-fighting mentality. All time goes to taking care of issues that are considered acute, leaving no time for developing activities or even normal daily work.
Personnel feel they have too much work and report stress. Sick leaves are higher than average in number and longer in duration than the average.
Organization is condescending to its negative conditions and does not seek to improve them.
There is low tolerance for uncertainty and a strong focus on finding permanent solutions to ambiguous issues. Discussions without immediate corrective actions are discouraged.
There is a fear of punishment if one reports a safety concern, human error, near-miss, or an incident. This leads to fewer reports of near misses compared to actual incidents.
The organization isolates itself from the environment, does not share information with others, or seek to learn from others. Organization's self-image can be overly positive.

Appendix B: Simplified assessment method

Table B. Simplified assessment of safety culture development stage (IAEA, 2002b, pp. 19-21).

	Stage I Rule-based	**Stage II** Goal-based	**Stage III** Improvement-based
Time focus	Short-term is all-important.	People are rewarded for exceeding goals, regardless of long-term consequences. Numerical targets are specified for safety.	Short-term performance is analyzed to improve longer-term performance. Longer-term focus with anticipation of consequences.
View of mistake	People are blamed for non-compliance with rules. Organizations react defensively to criticism rather than listening and learning.	Mistakes result in more controls and training.	Mistakes are an opportunity to understand and improve.
Handling of conflict	Conflicts are rarely resolved, and groups continue to compete with one another.	Conflict is discouraged in the name of teamwork.	Conflict is resolved by means of mutually beneficial solutions.
Role of managers	Managers enforce rules and pressure employees for results.	Managers use techniques such as management by objectives.	Managers coach people to improve performance. Managers support collaborative work.
View of people	People are components in a system.	Growing awareness that people's attitudes influence their performance.	People are respected and valued for their contribution.